

Statistical Physics of Fields

Course notes (following M. Kardar)

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1 Collective behaviour: from particles to fields

Statistical physics is the bridge from the microscopic to the macroscopic description:

micro	$\xrightarrow{\text{stat. mech.}}$	macro
μ -state: $\{p_i, q_i\}, \vec{s}_i, n_i, \dots$	$p_\mu \propto e^{-H(\mu)/k_B T}$	M-state: P, V, T
dynamics: $H(\mu)$	$N \gg 1$	thermodynamics

Interactions lead to **collective behaviour** (e.g. liquid crystals, superconductors). Mathematics forces things to simplify when many degrees of freedom are put together.

1.1 Phonons and elasticity

A solid is a perfect crystal plus deformations. At $T = 0$ the ions sit at $\vec{q}_\vec{r}^* = l\vec{a} + m\vec{b} + n\vec{c}$, with (l, m, n) labelling the lattice site; at finite temperature $\vec{q}_\vec{r} = \vec{q}_\vec{r}^* + \vec{u}(\vec{r})$. Then

$$H = \sum_{\vec{r}} \frac{\vec{p}_{\vec{r}}^2}{2m} + \mathcal{U}(\vec{q}_1, \dots, \vec{q}_N), \quad \mathcal{U} \simeq \mathcal{U}_0 + \frac{1}{2} \sum_{\vec{r}, \vec{r}'} \sum_{\alpha, \beta} \underbrace{\frac{\partial^2 \mathcal{U}}{\partial q_{\vec{r}, \alpha}^* \partial q_{\vec{r}', \beta}^*}}_{K_{\alpha\beta}(\vec{r} - \vec{r}')} u_\alpha(\vec{r}) u_\beta(\vec{r}') + \mathcal{O}(u^3),$$

with K a $3N \times 3N$ matrix. Normal modes follow by Fourier transformation, $\tilde{K}_{\alpha\beta}(\vec{k}) = \sum_{\vec{r}} K_{\alpha\beta}(\vec{r}) e^{i\vec{k} \cdot \vec{r}}$:

$$\mathcal{U} - \mathcal{U}_0 = \frac{1}{2} \sum_{\vec{r}, \vec{r}', \alpha, \beta} K_{\alpha\beta}(\vec{r} - \vec{r}') u_\alpha(\vec{r}) u_\beta(\vec{r}') = \frac{1}{2} \sum_{\vec{k}, \alpha, \beta} \tilde{K}_{\alpha\beta}(\vec{k}) \tilde{u}_\alpha(\vec{k}) \tilde{u}_\beta^*(\vec{k}).$$

Diagonalising the 3×3 matrix $\tilde{K}(\vec{k})$ gives eigenvalues $\kappa_\alpha(\vec{k})$ and eigenfrequencies $\omega_\alpha(\vec{k}) = \sqrt{\kappa_\alpha(\vec{k})/m}$, and

$$E(T) - \mathcal{U}_0 \propto \sum_{\vec{k}, \alpha} \frac{\hbar \omega_\alpha(\vec{k})}{e^{\beta \hbar \omega_\alpha(\vec{k})} - 1}.$$

One-dimensional chain. With lattice spacing a and displacements u_n ,

$$\mathcal{U} - \mathcal{U}_0 = \sum_n \left[\frac{K_1}{2} (u_{n+1} - u_n)^2 + \frac{K_2}{2} (u_{n+2} - u_n)^2 + \dots \right].$$

With $u_n = \sum_k e^{ikna} \tilde{u}(k)$, the sum over n gives $\delta_{k+k', 0}$ and

$$\mathcal{U} - \mathcal{U}_0 \propto \frac{1}{2} \sum_k |\tilde{u}(k)|^2 [K_1(2 - 2 \cos ka) + K_2(2 - 2 \cos 2ka) + \dots],$$

$$\omega(k) = \sqrt{\frac{K_1(2 - 2 \cos ka) + K_2(2 - 2 \cos 2ka) + \dots}{m}}.$$

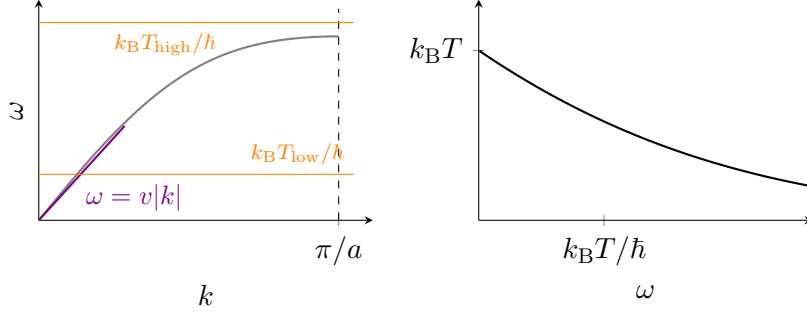


Figure 1.1: Left: dispersion of the chain over the Brillouin zone (π/a is the shortest length scale); translation symmetry forces $\omega \rightarrow 0$ as $k \rightarrow 0$. Right: the thermal weight $\hbar\omega/(e^{\beta\hbar\omega} - 1)$ cuts off modes with $\hbar\omega \gtrsim k_B T$.

At low temperature only the linear part matters:

$$E(T) - E_0 \xrightarrow{T \rightarrow 0} \int_0^\infty dk \frac{\hbar v |k|}{e^{\beta \hbar v |k|} - 1} \propto (k_B T)^2 \implies C(T) = \frac{dE}{dT} \propto T,$$

a quite general result.

1.2 Coarse graining

Average u over a window dx with $\lambda \gg dx \gg a$ (a “moving average”), where $\lambda \propto 1/T \gg a$ is the typical wavelength of the thermally excited modes. Construct $\mathcal{U}[u(x)]$ from:

- **locality:** $\mathcal{U}[u(x)] = \int dx \phi[u(x), du/dx, d^2u/dx^2, \dots]$ (higher derivatives encode interactions between more distant neighbours);
- **symmetry:** $\mathcal{U}[u(x) + c] = \mathcal{U}[u(x)]$ (uniform translation), so ϕ does not depend on u itself;
- **stability:** quadratic in the derivatives.

Thus

$$\mathcal{U}[u(x)] = \int dx \left[\frac{K}{2} \left(\frac{du}{dx} \right)^2 + \frac{L}{2} \left(\frac{d^2u}{dx^2} \right)^2 + \dots + M \left(\frac{d^2u}{dx^2} \right) \left(\frac{du}{dx} \right)^2 + \dots \right] \\ \xrightarrow{\text{F.T.}} \sum_k \left[\frac{K}{2} k^2 + \frac{L}{2} k^4 + \dots \right] |\tilde{u}(k)|^2.$$

Elastic solid in d dimensions. For $\Delta\mathcal{U} = \sum_{\vec{r}, \vec{r}'} \sum_{\alpha\beta} u_\alpha(\vec{r}) u_\beta(\vec{r}') K_{\alpha\beta}(\vec{r} - \vec{r}')$, $\Delta\mathcal{U} = \sum_{\vec{k}} \tilde{u}_\alpha(\vec{k}) \tilde{u}_\beta^*(\vec{k}) \tilde{K}_{\alpha\beta}(\vec{k})$, with $\tilde{K}_{\alpha\beta}(\vec{k} = 0) = 0$ by translation symmetry. By locality and isotropy,

$$\tilde{K}_{\alpha\beta}(\vec{k}) = \frac{\mu}{2} k^2 \delta_{\alpha\beta} + \frac{\mu + \lambda}{2} k_\alpha k_\beta + \mathcal{O}(k^4)$$

(an anisotropic material would allow other combinations such as $\#k_\alpha^2 + \#k_\beta^2$), so for an isotropic material

$$\beta\mathcal{H} = \sum_{\vec{k}} \left[\frac{\mu}{2} k^2 |\tilde{\vec{u}}|^2 + \frac{\mu + \lambda}{2} |\vec{k} \cdot \tilde{\vec{u}}|^2 \right] :$$

transverse modes ($\tilde{\vec{u}} \perp \vec{k}$) feel only μ , longitudinal ones $2\mu + \lambda$; in 3D there are two transverse branches and one longitudinal. With $\omega_{\max} \simeq k_B T / \hbar$,

$$C \stackrel{\text{low } T}{\propto} k_B V \left(\frac{k_B T}{\hbar v} \right)^d \sim A(v_l, v_t, \dots) T^d :$$

universal behaviour.

1.3 Phase transitions

Liquid–gas. The phase diagram of water (ice, liquid, steam) has a liquid–gas coexistence line ending at a critical point (T_c, P_c) , where critical opalescence is observed. Isotherms show a discontinuity (coexistence region) for $T < T_c$. The behaviour near T_c is universal, provided there are a dense and a dilute phase. Microscopically,

$$Z = \frac{1}{N!} \int \prod_{i=1}^N \frac{d^3 q_i}{\lambda^3} e^{-\beta \sum_{i<j} \mathcal{U}(\vec{q}_i - \vec{q}_j)}.$$

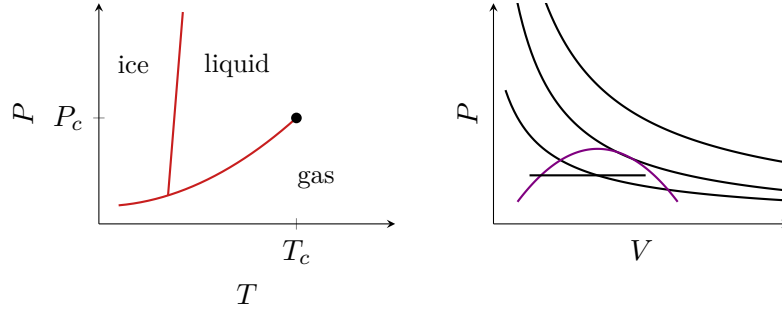


Figure 1.2: Liquid–gas transition: phase diagram with the critical point, and isotherms with the coexistence region below T_c .

Ferromagnet. In a field h , the magnetisation $m(h)$ is smooth for $T > T_c$ (paramagnet) and discontinuous at $h = 0$ for $T < T_c$ (ferromagnet), as for the liquid–gas transition. With $t = (T_c - T)/T_c$ in absolute value:

- $m(T \rightarrow T_c^-, h = 0) \propto |t|^\beta$; $m(T = T_c, h \rightarrow 0) \propto h^{1/\delta}$;
- susceptibility (compressibility for the liquid–gas): $\chi_\pm(h = 0) \propto |t|^{-\gamma_\pm}$, $\gamma_+ = \gamma_- = \gamma$, divergent at T_c (collective behaviour);
- heat capacity: $C_\pm \propto |t|^{-\alpha_\pm}$, $\alpha_+ = \alpha_- = \alpha$.

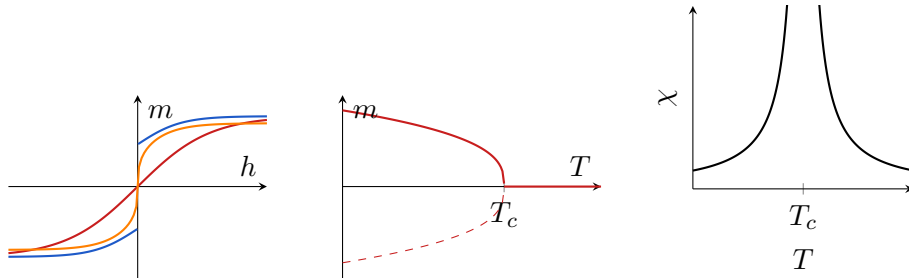


Figure 1.3: Magnetisation isotherms ($T > T_c$ red, $T = T_c$ orange, $T < T_c$ blue), spontaneous magnetisation at $h = 0$ (the sign depends on history), and the diverging susceptibility.

Susceptibility and correlations. In the Gibbs ensemble $Z(T, \vec{h}) = \text{Tr}[e^{-\beta H + \beta \vec{h} \cdot \vec{M}}]$; for $\vec{h} \parallel \vec{M}$,

$$\langle M \rangle = \frac{\partial \ln Z}{\partial (\beta h)} = \frac{1}{Z} \text{Tr}[M e^{-\beta H + \beta h M}],$$

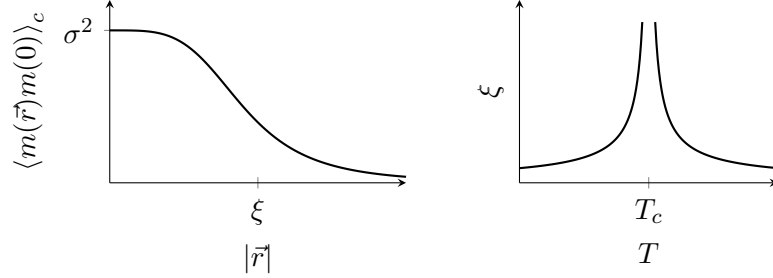
$$\chi := \frac{\partial \langle M \rangle}{\partial h} = \beta \left[\frac{\text{Tr}[M^2 e^{\dots}]}{Z} - \frac{(\text{Tr}[M e^{\dots}])^2}{Z^2} \right] = \beta [\langle M^2 \rangle - \langle M \rangle^2].$$

With $M = \int d^d r m(\vec{r})$,

$$\chi = \beta \int d^d r d^d r' [\langle m(\vec{r})m(\vec{r}') \rangle - \langle m(\vec{r}) \rangle \langle m(\vec{r}') \rangle] = \beta V \int d^d r \langle m(\vec{r})m(\vec{0}) \rangle_c,$$

since the connected correlation (covariance) depends only on the separation. If $\langle m(\vec{r})m(\vec{0}) \rangle_c \leq \sigma^2$ and decays beyond the **correlation length** ξ ,

$$\frac{\chi}{V} \lesssim \beta \sigma^2 \xi^d, \quad \frac{\chi}{V} \rightarrow \infty \iff \xi \rightarrow \infty, \quad \xi \propto |t|^{-\nu}.$$



2 The Landau–Ginzburg approach

2.1 Statistical field theory

Introduce a coarse-grained local magnetisation $\vec{m}(\vec{r})$, with $\vec{r} = (x_1, \dots, x_d)$ and $\vec{m} = (m_1, \dots, m_n)$ (liquid–gas: $n = 1$; superfluidity: $n = 2$; magnet: $n = 3$). Its probability is written as

$$\ln W[\vec{m}(\vec{r})] := -\beta \mathcal{H}[\vec{m}(\vec{r})],$$

where $\mathcal{H}$ is an *effective* Hamiltonian, different from the true microscopic one: it is a way of describing the logarithm of the probability of the different configurations. In principle W follows from a change of variables in $e^{-\beta H_{\text{mic}}}$; in practice it is constructed from symmetries and a few general principles:

- **locality**: $\beta \mathcal{H} = \int d^d x \phi[\vec{m}(\vec{x}), \nabla \vec{m}, \nabla^2 \vec{m}, \dots]$;
- **uniformity**: no explicit dependence on $\vec{x}$;
- **rotational symmetry** for $\vec{h} = 0$: only $m^2 = \sum_{\alpha} m_{\alpha} m_{\alpha}$, $m^4 = (\vec{m} \cdot \vec{m})^2, \dots, (\nabla \vec{m})^2 = \sum_{\alpha, i} \partial_i m_{\alpha} \partial_i m_{\alpha}, \dots$ appear;
- **analyticity**: the coarse-grained distribution is a sum of many microscopic contributions, which by the central limit theorem makes it a soft, analytic function even if the microscopic distributions are singular.

Hence the **Landau–Ginzburg** form

$$Z(T, \vec{h}) = \text{Tr}[e^{-\beta H_{\text{mic}}}] = e^{-\beta F_0} \int \mathcal{D}\vec{m}(\vec{x}) e^{-\beta \mathcal{H}[\vec{m}]},$$

$$\beta \mathcal{H} = \int d^d x \left[\frac{t}{2} m^2 + u m^4 + v m^6 + \dots + \frac{K}{2} (\nabla \vec{m})^2 + \frac{L}{2} (\nabla^2 \vec{m})^2 + \dots - \vec{h} \cdot \vec{m} \right].$$

2.2 Saddle-point approximation

Neither the trace over microstates nor the functional integral can be done exactly, but the weight can be guessed (with analytic coefficients). The singularities only appear in the thermodynamic limit $V \rightarrow \infty$:

$$Z(T, h) = Z_{\text{reg}}(T) \int \mathcal{D}\vec{m} e^{-\beta \mathcal{H}} = e^{-V F_{\text{reg}}} e^{-V F_{\text{sing}}},$$

with t, u, K analytic functions of T (not necessarily $\propto 1/T$). For $K > 0$ the exponent is maximal for a uniform configuration $\vec{m}(\vec{x}) = \bar{m}\hat{h}$, and

$$F_{\text{sing}} = \min_m \Psi(m), \quad \Psi(m) = \frac{t}{2}m^2 + um^4 + \dots - hm :$$

Ψ is analytic, but taking the minimum introduces non-analyticity. Expanding near the critical temperature of the specific material, $t = a(T - T_c) + b(T - T_c)^2 + \dots$, $u = u_0 + u_1(T - T_c) + \dots$, $K = K_0 + K_1(T - T_c) + \dots$, with $a > 0$ (the system is paramagnetic above T_c), $u_0 > 0$ (stability) and $K_0 > 0$ (ferromagnetic behaviour).

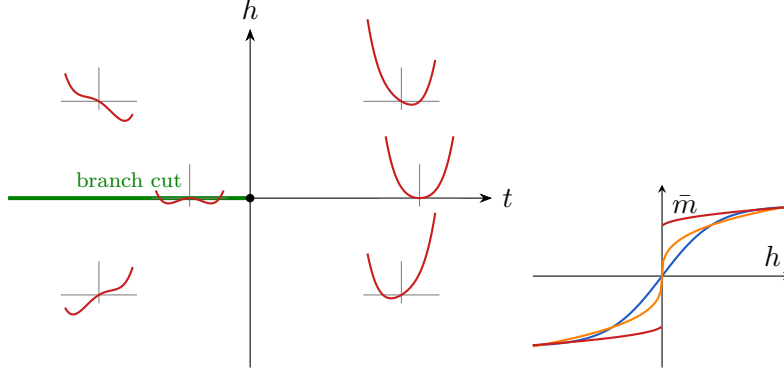


Figure 2.1: The Landau function $\Psi(m)$ in different regions of the (t, h) plane: a single minimum for $t > 0$, two minima for $t < 0$; the discontinuity of $\bar{m}$ along $t < 0, h = 0$ (branch cut). The simple theory reproduces the ferro-paramagnetic transition ($t < 0$ red, $t = 0$ orange, $t > 0$ blue).

Magnetisation. $d\Psi/dm = t\bar{m} + 4u\bar{m}^3 - h = 0$. For $h = 0$,

$$\bar{m} = \begin{cases} 0, & t > 0, \\ \pm\sqrt{-t/4u}, & t < 0, \end{cases}$$

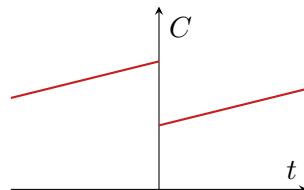
and for $t = 0$, $4u\bar{m}^3 = h$, i.e. $\bar{m} = (h/4u)^{1/3}$: $\beta = 1/2$, $\delta = 3$.

Susceptibility. $\chi = d\bar{m}/dh|_{h=0}$ with $h = t\bar{m} + 4u\bar{m}^3$, so $dh/d\bar{m} = t + 12u\bar{m}^2$, which equals t for $t > 0$ and $-2t$ for $t < 0$: $\gamma_+ = \gamma_- = \gamma = 1$, and with $\chi_{\pm} \simeq A_{\pm}|T - T_c|^{-\gamma_{\pm}}$ the amplitude ratio $A_+/A_- = 2$ is universal.

Heat capacity. $C(T, h = 0) = dE/dT = -\frac{d}{dT} \frac{\partial}{\partial \beta} \ln Z$; near T_c , $-\frac{\partial}{\partial T} \frac{\partial}{\partial \beta} \simeq k_B T_c^2 \frac{\partial^2}{\partial T^2} = a^2 k_B T_c^2 \frac{\partial^2}{\partial t^2}$, and

$$\ln Z = \ln Z_{\text{reg}} - V \left(\frac{t}{2} \bar{m}^2 + u \bar{m}^4 \right) = \ln Z_{\text{reg}}(t) + V \begin{cases} 0, & t > 0, \\ \frac{t^2}{16u}, & t < 0. \end{cases}$$

The heat capacity has a jump but no divergence: $\alpha = 0$ (discontinuity).



Superconductors have these exponents, but other systems do not. This is a problem, which probably comes from the saddle-point approximation.

2.3 Spontaneous symmetry breaking and Goldstone modes

Even for $\vec{h} = 0$ there are situations where $\bar{m} \neq 0$: the symmetry of the theory is spontaneously broken. The whole system can be rotated and remain a valid ground state, so slow deformations (wavelength λ of the order of the system size) cost little energy. These low-energy, long-wavelength modes are the **Goldstone modes** of a broken continuous symmetry; phonons are an example.

Superfluid. With $\psi(\vec{x}) = \psi_1 + i\psi_2$ (overlap of the wave function with the ground state at $\vec{x}$; $n = 2$),

$$\beta\mathcal{H}[\psi] = \int d^d x \left[\frac{t}{2} |\psi|^2 + u |\psi|^4 + \dots + \frac{K}{2} |\nabla \psi|^2 \right].$$

For $t < 0$ the minima form a circle; the cost of a phase deformation $\psi(\vec{x}) = \bar{\psi} e^{i\theta(\vec{x})}$ (“superfluid flow”) is

$$\beta\mathcal{H}[\bar{\psi} e^{i\theta(\vec{x})}] = \beta\mathcal{H}_0 + \frac{\bar{K}}{2} \int d^d x (\nabla \theta)^2, \quad \bar{K} = K \bar{\psi}^2.$$

With $\theta(\vec{x}) = \sum_{\vec{q}} e^{i\vec{q} \cdot \vec{x}} \tilde{\theta}_{\vec{q}} / \sqrt{V}$,

$$\beta\mathcal{H} = \beta\mathcal{H}_0 + \sum_{\vec{q}} \frac{\bar{K}}{2} q^2 |\tilde{\theta}_{\vec{q}}|^2, \quad \mathcal{P}[\tilde{\theta}_{\vec{q}}] \propto \prod_{\vec{q}} e^{-\frac{\bar{K}}{2} q^2 |\tilde{\theta}_{\vec{q}}|^2}, \quad \langle \tilde{\theta}_{\vec{q}} \rangle = 0, \quad \langle |\tilde{\theta}_{\vec{q}}|^2 \rangle = \frac{1}{\bar{K} q^2} :$$

long-wavelength fluctuations become larger because their energy cost is smaller. Then

$$\langle \theta(\vec{x}) \theta(\vec{x}') \rangle = \sum_{\vec{q}, \vec{q}'} \frac{e^{i\vec{q} \cdot \vec{x} + i\vec{q}' \cdot \vec{x}'}}{V} \underbrace{\langle \tilde{\theta}_{\vec{q}} \tilde{\theta}_{\vec{q}'} \rangle}_{\delta_{\vec{q}+\vec{q}', 0}/(\bar{K} q^2)} = \frac{1}{\bar{K}} \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot (\vec{x} - \vec{x}')}}{q^2} = -\frac{C_d(\vec{x} - \vec{x}')}{\bar{K}},$$

where $C_d(\vec{x}) := -\int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot \vec{x}}}{q^2}$ is the Coulomb potential, $\nabla^2 C_d = \delta^d(\vec{x})$. By Gauss’s theorem on a sphere of radius x , $1 = \oint d\vec{S} \cdot \nabla C_d = |dC_d/dx| x^{d-1} S_d$, with $S_d = 2\pi^{d/2}/(d/2 - 1)!$, so

$$C_d(x) = \frac{x^{2-d}}{(2-d)S_d} + C_0 \xrightarrow{|x| \rightarrow \infty} \begin{cases} C_0, & d > 2, \\ \frac{x^{2-d}}{(2-d)S_d}, & d < 2, \\ \frac{1}{2\pi} \ln x + C_0, & d = 2. \end{cases}$$

Therefore, with a the lattice spacing,

$$\langle [\theta(\vec{x}) - \theta(\vec{x}')]^2 \rangle = 2 \langle \theta^2 \rangle - 2 \langle \theta(\vec{x}) \theta(\vec{x}') \rangle = \frac{2(x^{2-d} - a^{2-d})}{\bar{K}(2-d)S_d} \xrightarrow{x \rightarrow \infty} \begin{cases} \text{const}, & d > 2, \\ \infty, & d \leq 2. \end{cases}$$

Fluctuations destroy continuous long-range order in $d \leq 2$: the **Mermin–Wagner theorem**; $d = 2$ is the lower critical dimension for continuous symmetries.

2.4 Fluctuations and scattering

In a scattering experiment an incoming wave $\vec{k}_i$ is elastically scattered to $\vec{k}_f$ with momentum transfer $|\vec{q}| = 2k \sin(\theta/2)$. Near T_c (small magnetisation, $t \propto T - T_c$) the configurations are weighted by

$$\mathcal{P}[\vec{m}(\vec{x})] \propto \exp \left\{ - \int d^d x \left[\frac{t}{2} m^2 + u m^4 + \dots + \frac{K}{2} (\nabla \vec{m})^2 + \dots \right] \right\}.$$

The scattering amplitude is $A(\vec{q}) \propto \langle \vec{k}_f | \sum_{\alpha} U(\vec{x}_{\alpha}) | \vec{k}_i \rangle \propto \sigma(\vec{q}) \int d^d x \rho(\vec{x}) e^{i\vec{q} \cdot \vec{x}} = \sigma(\vec{q}) \tilde{\rho}(\vec{q})$, and the intensity $S(\vec{q}) \propto |A(\vec{q})|^2 \propto \langle |\tilde{\rho}(\vec{q})|^2 \rangle$, which probes the fluctuations at small q .

Fluctuations around the most probable configuration $\vec{m}(\vec{x}) = \bar{m} \hat{e}_{\ell}$ ($\bar{m} = 0$ for $t > 0$, $\sqrt{-t/4u}$ for $t < 0$): write

$$\vec{m}(\vec{x}) = (\bar{m} + \phi_{\ell}(\vec{x})) \hat{e}_{\ell} + \sum_{\alpha=2}^n \phi_{t,\alpha}(\vec{x}) \hat{e}_{\alpha},$$

with a longitudinal and $n - 1$ transverse components. Then $(\nabla \vec{m})^2 = (\nabla \phi_{\ell})^2 + (\nabla \vec{\phi}_t)^2$, $m^2 = \bar{m}^2 + 2\bar{m}\phi_{\ell} + \phi_{\ell}^2 + \vec{\phi}_t^2$, $m^4 = \bar{m}^4 + 4\bar{m}^3\phi_{\ell} + 6\bar{m}^2\phi_{\ell}^2 + 2\bar{m}^2\vec{\phi}_t^2 + \mathcal{O}(\phi^3)$, so

$$\mathcal{P}[\phi_{\ell}, \vec{\phi}_t] \propto \exp \left\{ - \int d^d x \left[\frac{K}{2} (\nabla \phi_{\ell})^2 + \frac{t + 12u\bar{m}^2}{2} \phi_{\ell}^2 + \frac{K}{2} (\nabla \vec{\phi}_t)^2 + \frac{t + 4u\bar{m}^2}{2} \vec{\phi}_t^2 + \dots \right] \right\}.$$

The linear term $\phi_{\ell}(t\bar{m} + 4u\bar{m}^3)$ vanishes for the most probable configuration. With $\phi_{\alpha}(\vec{x}) = \sum_{\vec{q}} e^{i\vec{q} \cdot \vec{x}} \tilde{\phi}_{\alpha}(\vec{q}) / \sqrt{V}$,

$$\mathcal{P}[\tilde{\phi}_{\ell}(\vec{q}), \tilde{\phi}_t(\vec{q})] \propto \prod_{\vec{q}} e^{-\frac{K}{2}(q^2 + \xi_{\ell}^{-2}) |\tilde{\phi}_{\ell}(\vec{q})|^2} e^{-\frac{K}{2}(q^2 + \xi_t^{-2}) |\tilde{\phi}_t(\vec{q})|^2},$$

with

$$\frac{K}{\xi_{\ell}^2} := t + 12u\bar{m}^2 = \begin{cases} t, & t > 0, \\ -2t, & t < 0, \end{cases} = \chi_{\ell}^{-1}, \quad \frac{K}{\xi_t^2} := t + 4u\bar{m}^2 = \begin{cases} t, & t > 0, \\ 0, & t < 0, \end{cases} = \chi_t^{-1}.$$

The vanishing transverse “mass” for $t < 0$ corresponds to the Goldstone modes. The correlations are Lorentzian:

$$\langle \tilde{\phi}_{\alpha}(\vec{q}) \tilde{\phi}_{\beta}^*(\vec{q}') \rangle = \frac{\delta_{\vec{q}, \vec{q}'} \delta_{\alpha\beta}}{K(q^2 + \xi_{\alpha}^{-2})}.$$

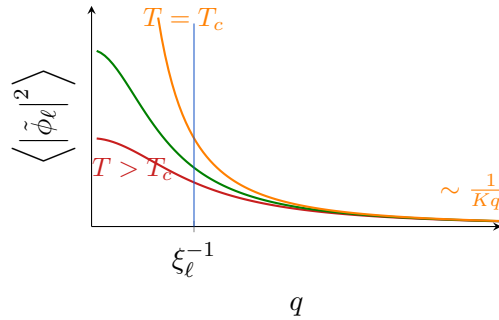


Figure 2.2: Lorentzian structure factor; at $T = T_c$ it diverges as $1/(Kq^2)$.

Experiments at T_c ($t = 0$) find instead $S(q) \propto 1/|q|^{2-\eta}$, with a new exponent η . In real space,

$$\langle \phi_{\ell}(\vec{x}) \phi_{\ell}(\vec{x}') \rangle = \sum_{\vec{q}, \vec{q}'} \frac{e^{i\vec{q} \cdot \vec{x} + i\vec{q}' \cdot \vec{x}'}}{V} \langle \tilde{\phi}_{\ell}(\vec{q}) \tilde{\phi}_{\ell}(\vec{q}') \rangle = \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot (\vec{x} - \vec{x}')}}{K(q^2 + \xi_{\ell}^{-2})} := -\frac{1}{K} I_d(\vec{x} - \vec{x}'; \xi_{\ell}).$$

The screened Coulomb potential. $I_d(\vec{x}; \xi) := - \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot \vec{x}}}{q^2 + \xi^{-2}}$ satisfies

$$\nabla^2 I_d = \int \frac{d^d q}{(2\pi)^d} \frac{(q^2 + \xi^{-2}) - \xi^{-2}}{q^2 + \xi^{-2}} e^{i\vec{q} \cdot \vec{x}} = \delta^d(\vec{x}) + \frac{I_d(\vec{x})}{\xi^2},$$

with a damping term. By spherical symmetry,

$$\frac{1}{x^{d-1}} \frac{d}{dx} \left(x^{d-1} \frac{dI_d}{dx} \right) = \frac{d^2 I_d}{dx^2} + \frac{d-1}{x} \frac{dI_d}{dx} = \frac{I_d}{\xi^2} + \delta^d(\vec{x}).$$

In $d = 1$, $I_1 \propto e^{-|x|/\xi}$. In general try $I_d \propto e^{-|x|/\xi}/x^p$ (though one needs to be careful): $dI_d/dx = -(\frac{1}{\xi} + \frac{p}{x})I_d$, $d^2 I_d/dx^2 = (\frac{1}{\xi^2} + \frac{2p}{\xi x} + \frac{p(p+1)}{x^2})I_d$, so for $x \neq 0$

$$\frac{2p}{\xi x} + \frac{p(p+1)}{x^2} - \frac{d-1}{\xi x} - \frac{(d-1)p}{x^2} = 0.$$

For $x \ll \xi$: $p(p+1) = p(d-1)$, so $p = d-2$ (at short distances the potential is not yet screened: standard Coulomb potential). For $x \gg \xi$: $2p = d-1$, so $p = (d-1)/2$:

$$-I_d(x) \simeq \begin{cases} \frac{x^{2-d}}{(d-2)S_d}, & x \ll \xi, \\ \frac{\xi^{(3-d)/2}}{(d-2)S_d} \frac{e^{-x/\xi}}{x^{(d-1)/2}}, & x \gg \xi, \end{cases}$$

the prefactor of the second form chosen to match dimensions and the first form at $x \sim \xi$.

Correlation length. $\xi_\ell = \sqrt{K/t}$ for $t > 0$ and $\sqrt{K/(-2t)}$ for $t < 0$, i.e. $\xi_\ell = B_\pm |T - T_c|^{-\nu_\pm}$ with $\nu_+ = \nu_- = \nu = 1/2$ and $B_+/B_- = \sqrt{2}$. The Goldstone modes have long-range correlations: $\xi_t \rightarrow \infty$ for $t < 0$.

Susceptibility. With $\langle \phi(\vec{x})\phi(\vec{0}) \rangle \propto x^{2-d-\eta}$ up to ξ ,

$$\chi \propto \int d^d x \langle \phi(\vec{x})\phi(\vec{0}) \rangle \propto \int^\xi d^d x x^{2-d-\eta} \propto \xi^{2-\eta} \propto |t|^{-\nu(2-\eta)} \implies \gamma = (2-\eta)\nu :$$

the divergence of the susceptibility is related to that of the correlation length.

transition	material	α	β	γ	ν
ferro/paramagnet ($n = 3$)	Fe, Ni	-0.1	0.4	1.3	0.7
superfluidity ($n = 2$)	He	0 (“ λ ”)	0.32	—	0.67
liquid/gas ($n = 1$)	CO ₂ , Ne	0.1	0.30	1.24	0.63
superconductors ($n = 2$)	Al, Cu	0 (disc.)	1/2	1	1/2

Superconductors and superfluids share the same symmetry ($n = 2$, $d = 3$) but not the same exponents; only superconductors show the saddle-point values.

2.5 Beyond the saddle point

Heat capacity. For liquid-gas ($n = 1$) $\alpha \simeq 0.11$; superfluid ($n = 2$) $\alpha \simeq 0$ (“ $\ln$ ”); ferromagnet ($n = 3$) $\alpha \simeq -0.1$, with $C \sim |t|^{-\alpha}$. In the saddle-point approximation (the most probable state),

$$Z_{\text{sing}}(t, h = 0) = e^{-V \cdot \{0; -t^2/16u\}}, \quad C_{\text{s.p.}}(t, h = 0) = \begin{cases} 0, & t > 0, \\ 1/8u, & t < 0, \end{cases}$$

i.e. $\alpha = 0$ (discontinuity), observed for superconductors.

Saddle-point method. For N large,

$$\begin{aligned}
I &= \int dm e^{-N\Psi(m)} = \int dm e^{-N[\Psi(\bar{m}) + \frac{1}{2}\Psi''(\bar{m})(m-\bar{m})^2 + \dots]} \\
&= e^{-N\Psi(\bar{m})} \int d\delta m e^{-\frac{N}{2}\Psi''(\bar{m})\delta m^2} [1 + N\mathcal{O}(\delta m^3) + \dots] \\
&= e^{-N\Psi(\bar{m})} \sqrt{\frac{2\pi}{N\Psi''(\bar{m})}} \left[1 + \mathcal{O}\left(\frac{1}{N}\right)\right], \\
\ln I &\simeq -N\Psi(\bar{m}) - \frac{1}{2} \ln \frac{N\Psi''(\bar{m})}{2\pi} + \mathcal{O}(1/N),
\end{aligned}$$

where the second term is negligible for $N \gg 1$. For the field theory, expanding around the saddle point,

$$\begin{aligned}
Z_{\text{sing}} &= e^{-V(\frac{t}{2}\bar{m}^2 + u\bar{m}^4)} \int \mathcal{D}\phi_\ell \mathcal{D}\vec{\phi}_t \\
&\quad \times e^{-\frac{K}{2} \int d^d x [\xi_\ell^{-2} \phi_\ell^2 + (\nabla \phi_\ell)^2]} e^{-\frac{K}{2} \int d^d x [\xi_t^{-2} \phi_t^2 + (\nabla \phi_t)^2]} [1 + \dots] :
\end{aligned}$$

the leading correction comes from the quadratic term. In Fourier space,

$$\begin{aligned}
Z_{\text{sing}} &= e^{-V(\frac{t}{2}\bar{m}^2 + u\bar{m}^4)} \prod_{\vec{q}} \int d\tilde{\phi}_\ell d\tilde{\phi}_t e^{-\frac{K}{2}(q^2 + \xi_\ell^{-2})|\tilde{\phi}_\ell|^2} e^{-\frac{K}{2}(q^2 + \xi_t^{-2})|\tilde{\phi}_t|^2} \\
&= e^{-V(\frac{t}{2}\bar{m}^2 + u\bar{m}^4)} \prod_{\vec{q}} \left(\frac{2\pi}{K(q^2 + \xi_\ell^{-2})} \right)^{1/2} \left(\frac{2\pi}{K(q^2 + \xi_t^{-2})} \right)^{(n-1)/2} [1 + \dots],
\end{aligned}$$

and with $\sum_{\vec{q}} \rightarrow V \int \frac{d^d q}{(2\pi)^d}$,

$$\begin{aligned}
-\frac{\ln Z_{\text{sing}}}{V} &= \left(\frac{t}{2}\bar{m}^2 + u\bar{m}^4 \right) \\
&\quad + \frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \ln(Kq^2 + K\xi_\ell^{-2}) + \frac{n-1}{2} \int \frac{d^d q}{(2\pi)^d} \ln(Kq^2 + K\xi_t^{-2}).
\end{aligned}$$

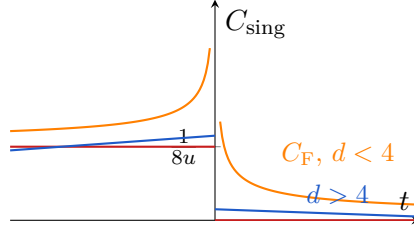
This is mathematically ill defined, but physically the field has a limited number of Fourier components, up to $\Lambda \propto 1/a$. Then

$$C \propto -\frac{\partial^2 \ln Z_{\text{sing}}}{\partial t^2} = \begin{cases} 0 + \frac{n}{2} \int \frac{d^d q}{(2\pi)^d} \frac{1}{(Kq^2 + t)^2}, & t > 0, \\ \frac{1}{8u} + 2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{(Kq^2 - 2t)^2}, & t < 0, \end{cases} = C_{\text{s.p.}} + C_{\text{F}}.$$

Are the corrections small?

$$\begin{aligned}
C_{\text{F}} &\propto \frac{1}{K^2} \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + \xi_\ell^{-2})^2} = \frac{1}{K^2} \frac{S_d}{(2\pi)^d} \int dq \frac{q^{d-1}}{(q^2 + \xi^{-2})^2}, \\
C_{\text{F}} &\propto \begin{cases} \frac{1}{K^2} \Lambda^{d-4}, & d > 4 \text{ (integral up to } \Lambda), \\ \frac{1}{K^2} \xi^{4-d} \propto t^{-\frac{4-d}{2}}, & d < 4 \text{ (integral to infinity)}. \end{cases}
\end{aligned}$$

For $d > 4$ the correction does not depend on t and the discontinuity survives; for $d < 4$ it diverges at T_c . The saddle-point method cannot be applied when the field fluctuates all over space.



Ginzburg criterion. Why does the saddle point work for superconductors? For $d < 4$, with $t = (T_c - T)/T_c$, $\xi \simeq \xi_0 t^{-1/2}$ and $\xi_0 \sim \sqrt{K}$, the fluctuation correction becomes comparable to the jump $\Delta C_{\text{s.p.}}$ when

$$\Delta C_{\text{s.p.}} \sim \frac{1}{K^2} \xi^{4-d} \sim \frac{1}{\xi_0^4} \xi_0^{4-d} t_G^{\frac{d-4}{2}} \implies t_G \sim (\Delta C_{\text{s.p.}} \xi_0^d)^{\frac{2}{d-4}} = \Delta C_{\text{s.p.}}^{-\frac{2}{4-d}} \xi_0^{-\frac{2d}{4-d}},$$

with $2d/(4-d) = 6$ in $d = 3$: the saddle point fails only for $|t| < t_G$. For the liquid–gas transition $\xi_0 \sim 1$ –10 atomic spacings; for superconductivity ξ_0 is the size of a Cooper pair, $\sim 10^3 \text{ \AA}$, so t_G is tiny: the resolution needed to see the effect of the fluctuations is too high. Another way of computing corrections is to start with $u = 0$ and expand: the corrections are of order $u \xi^{(4-d)/2}$.

3 The scaling hypothesis

Critical points (the end of the liquid–gas coexistence line, or T_c on the $h = 0$ axis) show **universality**: the exponents are independent of the material.

system	α	β	γ	δ	ν	η
liquid–gas ($n = 1$)	0.11	0.33	1.238	4.80	0.629	0.032
superfluid ($n = 2$)	−0.0127	0.35	1.32	4.79	0.67	0.04
ferromagnets ($n = 3$)	−0.12	0.37	1.39	4.79	0.71	0.04
saddle point	0 (disc.)	1/2	1	3	1/2	0
2D Ising ($d = 2, n = 1$)	0 (ln)	1/8	7/4	15	1	1/4

($\alpha, \beta, \gamma, \delta$ are macroscopic; ν, η microscopic, from light scattering; the superfluid α is a satellite measurement.)

3.1 Homogeneity of the free energy

Approaching the critical point along t or along h gives different singularities. What about other directions? In the saddle-point approximation, $f_{\text{sing}}(t, h) = \min_m [\frac{t}{2} m^2 + u m^4 + \dots - h m]$ with $m \propto \pm \sqrt{-t/4u}$ ($t < 0, h = 0$) and $(h/4u)^{1/3}$ ($t = 0$), so

$$f_{\text{sing}}(t, h) \propto \begin{cases} -t^2/u, & h = 0, \\ -h^{4/3}/u^{1/3}, & t = 0, \end{cases} \propto t^2 g_f\left(\frac{h}{t^\Delta}\right),$$

with $g_f(x \rightarrow 0) \rightarrow \text{const}$ and $g_f(x \rightarrow \infty) \propto x^p$. For $t \rightarrow 0$, $t^2(h/t^\Delta)^p \propto h^{4/3}$ requires $p = 4/3$ and $2 - \Delta p = 0$: the **gap exponent** is $\Delta = 3/2$ (the crossover happens along $h \sim t^{3/2}$).

Scaling hypothesis. Assume, for some α and Δ , the homogeneous form

$$f_{\text{sing}}(t, h) = |t|^{2-\alpha} g_f\left(\frac{h}{|t|^\Delta}\right).$$

Then $\partial f_{\text{sing}}/\partial t = t^{1-\alpha}[(2-\alpha)g_f - \Delta \frac{h}{t^\Delta} g'_f] = t^{1-\alpha}g_1(h/t^\Delta)$ and $C \propto \partial^2 f_{\text{sing}}/\partial t^2 \propto t^{-\alpha}g_2(h/t^\Delta)$. Going analytically around the critical point (through $h \neq 0$) from one side to the other, the only way the two functions can match is $\alpha_+ = \alpha_- = \alpha$.

Magnetisation. $m(t, h) = \partial f_{\text{sing}}/\partial h = t^{2-\alpha-\Delta}g'_f(h/t^\Delta)$: for $h \rightarrow 0$, $m \sim t^{2-\alpha-\Delta} = t^\beta$; for $t \rightarrow 0$, $m \sim t^{2-\alpha-\Delta}(h/t^\Delta)^{\bar{p}} \sim h^{1/\delta}$ requires $\bar{p} = (2 - \alpha - \Delta)/\Delta$:

$$\beta = 2 - \alpha - \Delta, \quad \delta = \frac{\Delta}{2 - \alpha - \Delta}.$$

Susceptibility. $\chi \propto \partial m/\partial h \propto t^{2-\alpha-2\Delta}g''_f(h/t^\Delta) \rightarrow t^{-\gamma}$, so $\gamma = 2\Delta - 2 + \alpha$.

Consequences: (i) the same critical exponents above and below T_c ; (ii) all quantities are homogeneous functions of h/t^Δ ; (iii) almost all exponents are related to α and Δ , giving **exponent identities**:

$$\begin{aligned} \alpha + 2\beta + \gamma &= 2 & (\text{Rushbrooke; } n = 3: -0.12 + 2 \cdot 0.37 + 1.39 = 2.01), \\ \delta - 1 &= \frac{\gamma}{\beta} & (\text{Widom; } n = 1: 4.8 - 1 = 3.8 \simeq 1.24/0.33 = 3.76). \end{aligned}$$

3.2 Homogeneity of the correlation length

ν and η are microscopic exponents; α, β, γ macroscopic. Assume $\xi(t, h) = t^{-\nu}g_\xi(h/t^\Delta)$, and that the divergence of ξ is solely responsible for the singular behaviour: dividing a system of size L into independent blocks of size ξ ,

$$\ln Z = \left(\frac{L}{a}\right)^d f_{\text{reg}}(t, h) + \left(\frac{L}{\xi}\right)^d \dots \implies \ln Z_{\text{sing}}(t, h) \propto \xi^{-d} \propto t^{d\nu} g_f\left(\frac{h}{t^\Delta}\right):$$

ξ is responsible for the change in the number of degrees of freedom. Comparing with $f_{\text{sing}} \propto t^{2-\alpha}$ gives the **Josephson (hyperscaling) identity**

$$d\nu = 2 - \alpha$$

($d = 3, n = 2: 3 \cdot 0.67 = 2.01 \simeq 2 + 0.01$; $d = 2, n = 1: 2 \cdot 1 = 2 - 0$). For the saddle point, $2 - 0 = d/2$ only holds for $d = 4$: fluctuations fix it.

At the critical point ($t = h = 0$) the only way correlations can decay is as a power law,

$$\langle m(\vec{x})m(\vec{y}) \rangle \sim \frac{1}{|\vec{x} - \vec{y}|^{d-2+\eta}}, \quad \chi \propto \int^\xi d^d x \langle m(\vec{x})m(\vec{0}) \rangle \propto \int^\xi \frac{d^d x}{x^{d-2+\eta}} \propto \xi^{2-\eta} \propto t^{-(2-\eta)\nu},$$

giving the **Fisher identity** $\gamma = (2 - \eta)\nu$ ($d = 2, n = 1: (2 - \frac{1}{4}) \cdot 1 = \frac{7}{4}$). At the critical point the correlation function is homogeneous, $G(\lambda|\vec{x} - \vec{y}|) = \lambda^p G(|\vec{x} - \vec{y}|)$: invariance under scaling, i.e. the configurations are *fractal*, with the same statistical properties (up to a factor λ^p) on all scales between a and L .

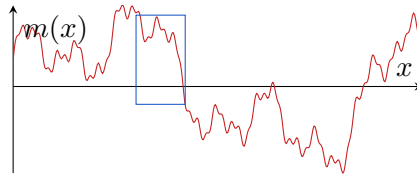


Figure 3.1: A self-similar configuration at the critical point: a zoomed window looks statistically the same after rescaling by λ^p .

Similarly, all response functions can be written as integrals of correlation functions (because they are second derivatives): with the energy density $E(\vec{x})$ and $\langle E(\vec{x})E(\vec{0}) \rangle_c \sim 1/|x|^{d-2+\eta'}$,

$$C_{\text{sing}}(t, h) \propto \int^\xi d^d x \langle E(\vec{x})E(\vec{0}) \rangle_c \propto \xi^{2-\eta'} \propto t^{-\nu(2-\eta')} \implies \alpha = \nu(2 - \eta').$$

In $d = 2$ scale invariance is related to conformal invariance: one can construct conformal theories and extract exponents. The effects of scale transformations on the system are the beginning of renormalisation.

3.3 Renormalisation

At $t = h = 0$, $f_{\text{sing}} = t^{2-\alpha} g_f(h/t^\Delta)$, $\xi = t^{-\nu} g_\xi(h/t^\Delta)$ and $\langle m(\vec{x})m(\vec{y}) \rangle \simeq 1/|\vec{x} - \vec{y}|^{d-2+\eta}$: **statistical self-similarity**, with an implicit shortest distance a . The **renormalisation group** (RG) transformation consists of:

- (i) **coarse grain** by averaging over a scale ba : $\tilde{m}(\vec{x}) = \frac{1}{(ba)^d} \int_{\text{box}} d^d y m(\vec{x} + \vec{y})$ (“bigger pixels”; set $a = 1$);
- (ii) **rescale**: $\vec{x}' = \vec{x}/b$;
- (iii) **renormalise**: $m' = \tilde{m}/\zeta$.

Mathematically, the transformation

$$m'(\vec{x}') = \frac{1}{\zeta b^d} \int_{\text{box}} d^d y m(b\vec{x}' + \vec{y})$$

is not invertible (one cannot go back), but it maps the old weight $\mathcal{P}_{\text{old}}[m(\vec{x})] = e^{-\beta\mathcal{H}[m]}$ onto a new one $\mathcal{P}_{\text{new}}[m'(\vec{x}')] = e^{-\beta\mathcal{H}'[m']}$, and the two are the same at the critical point ($t = h = 0$).

3.4 The renormalisation group (Kadanoff)

Away from the critical point the correlation length is reduced, $\xi(t, h) = b\xi(t', h')$, so the point (t, h) moves away from the origin. The mapping (which could be invertible) is

$$t'_b(t, h) = A(b)t + B(b)h + \mathcal{O}(t^2, h^2), \quad h'_b(t, h) = C(b)t + D(b)h + \mathcal{O}(t^2, h^2),$$

with no constant terms (the critical point maps onto itself), and $B = C = 0$ by symmetry ($h \rightarrow -h$). Composing two transformations, $A(b_1 b_2) = A(b_1)A(b_2)$ with $A(1) = 1$, so $A(b) = b^{y_t}$:

$$t' = b^{y_t} t + \dots, \quad h' = b^{y_h} h + \dots$$

Since $Z = \int \mathcal{D}m e^{-\beta\mathcal{H}} \rightarrow \int \mathcal{D}m' e^{-(\beta\mathcal{H})'}$ is unchanged, $\ln Z_{\text{sing}}(t, h) = \ln Z_{\text{sing}}(t', h')$, which measures how far one is from the critical point. With $V' = V/b^d$, $Vf(t, h) = V'f(t', h')$, so

$$f(t, h) = b^{-d} f(b^{y_t} t, b^{y_h} h),$$

a homogeneous function. Choosing b such that $b^{y_t} t = 1$, i.e. $b = t^{-1/y_t}$,

$$f(t, h) = t^{d/y_t} f\left(1, \frac{h}{t^{y_h/y_t}}\right), \quad \xi(t, h) = b\xi(b^{y_t} t, b^{y_h} h) = t^{-1/y_t} g_\xi\left(\frac{h}{t^{y_h/y_t}}\right),$$

which reproduces the scaling hypothesis with

$$2 - \alpha = \frac{d}{y_t}, \quad \Delta = \frac{y_h}{y_t}, \quad \nu = \frac{1}{y_t}.$$

For the magnetisation, $m(t, h) = \partial f / \partial h = b^{y_h-d} g(b^{y_t} t, b^{y_h} h) = t^{(d-y_h)/y_t} g(h/t^\Delta)$, so $\beta = (d - y_h)/y_t$, and the renormalisation factor is $\zeta = b^{d-y_h}$ (h and m are conjugate variables).

General procedure. Start from an experimental critical point and its statistical field $\vec{m}(\vec{x})$. Construct the most general weight consistent with locality, symmetry, stability, ...:

$$-\ln \mathcal{P}[\vec{m}(\vec{x})] = \beta \mathcal{H}[\vec{m}] = \int_a d^d x \left[\frac{t}{2} m^2 + u m^4 + v m^6 + \dots + \frac{K}{2} (\nabla m)^2 + \frac{L}{2} (\nabla^2 m)^2 + \dots \right],$$

with a the short-scale cut-off. Applying the three RG steps, $m'(\vec{x}') = \frac{1}{\zeta b^d} \int d^d y m(b\vec{x}' + \vec{y})$,

$$-\ln W[m'] = \text{const} + \int d^d x' \left[\frac{t'}{2} m'^2 + u' m'^4 + v' m'^6 + \dots + \frac{K'}{2} (\nabla m')^2 + \frac{L'}{2} (\nabla^2 m')^2 + \dots \right],$$

with $t'_b = t'_b(t, u, v, \dots)$, $u'_b = u'_b(t, u, v, \dots)$, etc.: a mapping $S' = \mathcal{R}_b[S]$ in the space of parameters $S = (t, u, v, K, \dots)$.

- Find the **fixed point**: $S^* = \mathcal{R}_b[S^*]$.
- Deviate from it: $\mathcal{R}_b[S^* + \delta S] = S^* + \delta S'$ with $\delta S' = \mathcal{R}_b^L \delta S$ (linearised transformation, a matrix).
- Diagonalise $\mathcal{R}^L$: $\mathcal{R}_b^L \hat{O}_i = \lambda_i(b) \hat{O}_i$. Since $\mathcal{R}_{b_1}^L \mathcal{R}_{b_2}^L = \mathcal{R}_{b_1 b_2}^L = \mathcal{R}_{b_2}^L \mathcal{R}_{b_1}^L$ they commute and share eigenvectors, and $\lambda_i(b_1) \lambda_i(b_2) = \lambda_i(b_1 b_2)$, $\lambda_i(1) = 1$, so $\lambda_i(b) = b^{y_i}$:

$$S = S^* + \sum_i a_i \hat{O}_i \longrightarrow S' = S^* + \sum_i a_i b^{y_i} \hat{O}_i.$$

If $y_i > 0$ the corresponding direction is **relevant**; if $y_i < 0$, **irrelevant**; if $y_i = 0$, **marginal** (higher orders are needed). The set of irrelevant directions of S^* defines its **basin of attraction**. Since $\xi(S') = \xi(S)/b$, at a fixed point $\xi(S^*) = \xi(S^*)/b$, so $\xi(S^*) = 0$ (disordered or ordered phases) or $\xi(S^*) \rightarrow \infty$ (critical fixed point); and for any point in the basin of attraction of a critical fixed point, $\xi \rightarrow \infty$ as well. After rescaling one sees the same thing once the irrelevant degrees of freedom have been washed out. (For glassy systems the RG trajectory can be chaotic.)

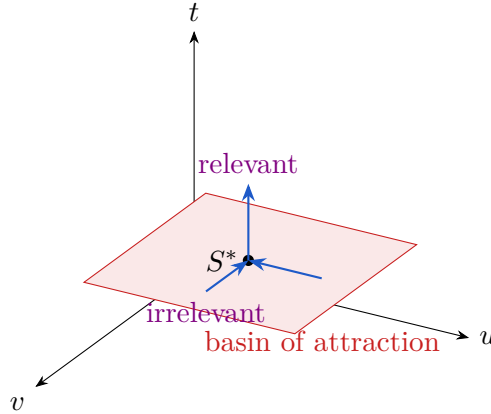


Figure 3.2: RG flow near a critical fixed point.

“All we are doing is mathematics, but at the end of the day we have to give it properties related to experiments” (M. Kardar).

3.5 The Gaussian model

Keeping only quadratic terms (then it is easy to solve),

$$Z = \int \mathcal{D}\vec{m}(\vec{x}) \exp \left\{ - \int_a d^d x \left[\frac{t}{2} m^2 + \frac{K}{2} (\nabla m)^2 + \frac{L}{2} (\nabla^2 m)^2 + \dots - \vec{h} \cdot \vec{m} \right] \right\},$$

with an implicit short scale a . The Fourier representation decouples the gradients:

$$\tilde{\vec{m}}(\vec{q}) = \int d^d x e^{i\vec{q}\cdot\vec{x}} \vec{m}(\vec{x}), \quad \vec{m}(\vec{x}) = \int \frac{d^d q}{(2\pi)^d} e^{-i\vec{q}\cdot\vec{x}} \tilde{\vec{m}}(\vec{q}) = \sum_{\vec{q}} \frac{e^{-i\vec{q}\cdot\vec{x}}}{V} \tilde{\vec{m}}(\vec{q}),$$

$$\begin{aligned} Z(t, h) &= \prod_{\vec{q}} \int \frac{d^n \tilde{\vec{m}}(\vec{q})}{V^{n/2}} \exp \left[-\frac{t + Kq^2 + Lq^4 + \dots}{2V} |\tilde{\vec{m}}(\vec{q})|^2 + \vec{h} \cdot \tilde{\vec{m}}(\vec{q}=0) \right] \\ &= \prod_{\vec{q}} \frac{1}{V^{n/2}} \left(\frac{2\pi V}{t + Kq^2 + Lq^4 + \dots} \right)^{n/2} e^{\frac{V}{2t} h^2}, \end{aligned}$$

$$f(t, h) = -\frac{\ln Z}{V} = \text{const} + \frac{n}{2} \int \frac{d^d q}{(2\pi)^d} \ln(t + Kq^2 + \dots) - \frac{h^2}{2t},$$

where the integral runs over the Brillouin zone ($|q_i| < \pi/a$), approximated by a sphere of radius Λ . The singular heat capacity ($t > 0$; otherwise the weight is ill defined) is

$$\begin{aligned} C_{\text{sing}}(t, h=0) &\sim \frac{\partial^2 f}{\partial t^2} = \frac{n}{2} \int^\Lambda \frac{d^d q}{(2\pi)^d} \frac{1}{(t + Kq^2 + Lq^4 + \dots)^2} \\ &\stackrel{Kq^2=tx^2}{=} \frac{n}{2} \frac{S_d}{(2\pi)^d} \left(\frac{t}{K} \right)^{d/2} \frac{1}{t^2} \int_0^{\sqrt{K/t}\Lambda} dx \frac{x^{d-1}}{(1+x^2+\dots)^2}, \end{aligned}$$

whose behaviour depends on the dimension. If $d < 4$ (keeping terms up to x^2) it is $\simeq At^{d/2-2}$, the singular term, so

$$f_{\text{sing}}(t, h) = t^{d/2} \left[B - \frac{h^2}{t^{1+d/2}} \right] \sim t^{2-\alpha} g_f \left(\frac{h}{t^\Delta} \right) \implies 2-\alpha = \frac{d}{2}, \quad \Delta = \frac{1}{2} + \frac{d}{4}.$$

The m^4 term is needed to study $t < 0$.

RG for the Gaussian model. The modes do not talk to each other, so:

- (i) **Coarse grain**, averaging from a to ba : $q'_{\text{max}} = q_{\text{max}}/b$; split $\tilde{\vec{m}}(\vec{q})$ into $\tilde{\vec{\sigma}}(\vec{q}^>)$ ($\Lambda/b < q < \Lambda$) and $\tilde{\vec{m}}(\vec{q}^<)$ ($q < \Lambda/b$):

$$\begin{aligned} Z &= \int \mathcal{D}\tilde{\vec{\sigma}}(\vec{q}^>) e^{-\int_{\Lambda/b}^\Lambda \frac{d^d q}{(2\pi)^d} \frac{t+Kq^2+\dots}{2} |\tilde{\vec{\sigma}}|^2} \\ &\quad \times \int \mathcal{D}\tilde{\vec{m}}(\vec{q}^<) e^{-\int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \left[\frac{t+Kq^2+\dots}{2} |\tilde{\vec{m}}|^2 + \vec{h} \cdot \tilde{\vec{m}}(0) \right]} \\ &= e^{-V\delta f_b(t)} \int \mathcal{D}\tilde{\vec{m}}(\vec{q}^<) e^{-\dots}. \end{aligned}$$

- (ii) & (iii) **Rescale and renormalise**: $\vec{x}' = \vec{x}/b$, $\vec{q}' = b\vec{q}$ (the integration goes back up to Λ), $m'(\vec{x}') = \tilde{m}(\vec{x})/\zeta$, i.e. $\tilde{m}'(\vec{q}') = \tilde{m}(\vec{q})/z$ with $z = b^{-d}\zeta$. Then

$$\begin{aligned} Z(t, h) &= e^{-V\delta f_b(t)} \int \mathcal{D}\tilde{m}'(\vec{q}') \exp \left\{ -\int_0^\Lambda \frac{d^d q'}{(2\pi)^d} \left[\frac{b^{-d}t + Kb^{-d-2}q'^2 + Lb^{-d-4}q'^4 + \dots}{2} z^2 |\tilde{m}'|^2 \right] + z\vec{h} \cdot \tilde{\vec{m}}'(0) \right\} \\ &= e^{-V\delta f_b(t)} Z(t', h'), \text{ with} \end{aligned}$$

$$t' = b^{-d}z^2t, \quad K' = b^{-d-2}z^2K, \quad L' = b^{-d-4}z^2L, \quad \dots, \quad h' = zh.$$

The physics comes in the choice of z . Choose z such that $K' = K$: $z = b^{(d+2)/2}$. Then

$$t' = b^2 t, \quad h' = b^{1+d/2} h \quad (\text{relevant: } y_t = 2, y_h = 1 + d/2), \quad L' = b^{-2} L, \dots \quad (\text{irrelevant}),$$

and $f_{\text{sing}}(t, h) = t^{d/y_t} g(h/t^{y_h/y_t}) = t^{d/2} g(h/t^{1/2+d/4})$. Choosing instead z such that $L' = L$ makes K, t, h all relevant; choosing $t' = t$ ($z = b^{d/2}$) gives $K' \rightarrow 0$, i.e. zero correlation length, with $\zeta = b^{-d/2}$: rediscovering the central limit theorem.

Relevance of the non-Gaussian terms. With $\vec{x} \rightarrow b\vec{x}'$, $m \rightarrow \zeta m'$ in $\beta\mathcal{H} = \int d^d x [\frac{t}{2} m^2 + um^4 + \dots + u_p m^p + \dots + \frac{K}{2} (\nabla m)^2 + \frac{L}{2} (\nabla^2 m)^2 + \dots]$:

$$t' = b^d \zeta^2 t, \quad u' = b^d \zeta^4 u, \quad \dots, \quad K' = b^{d-2} \zeta^2 K, \quad L' = b^{d-4} \zeta^2 L.$$

$K' = K$ requires $\zeta = b^{1-d/2}$, so $u'_p = b^d \zeta^p u_p = b^{d+p(1-d/2)} u_p = b^{y_p} u_p$, with

$$y_p = p + d\left(1 - \frac{p}{2}\right): \quad y_1 = 1 + \frac{d}{2} (= y_h), \quad y_2 = 2 (= y_t), \quad y_4 = 4 - d, \quad y_6 = 6 - 2d, \dots$$

Relevance depends on the dimension: in $d = 3$ the quartic term is relevant.

4 Perturbative renormalisation group

Split the weight into a Gaussian part and interactions (non-Gaussian terms):

$$Z = \int \mathcal{D}\vec{m}(\vec{x}) e^{-\beta\mathcal{H}[\vec{m}]} = \int \mathcal{D}\vec{m}(\vec{x}) e^{-(\beta\mathcal{H})_0 - \mathcal{U}},$$

$$(\beta\mathcal{H})_0 = \int d^d x \left[\frac{t}{2} m^2 + \frac{K}{2} (\nabla m)^2 + \frac{L}{2} (\nabla^2 m)^2 + \dots \right], \quad \mathcal{U} = \int d^d x [um^4 + vm^6 + \dots].$$

With $\vec{m}(\vec{x}) = \frac{1}{V} \sum_{\vec{q}} e^{i\vec{q}\cdot\vec{x}} \tilde{\vec{m}}(\vec{q}) = \int \frac{d^d q}{(2\pi)^d} e^{i\vec{q}\cdot\vec{x}} \tilde{\vec{m}}(\vec{q})$,

$$(\beta\mathcal{H})_0 = \frac{1}{V} \sum_{\vec{q}} \frac{1}{2} (t + Kq^2 + \dots) |\tilde{\vec{m}}(\vec{q})|^2 = \int \frac{d^d q}{(2\pi)^d} \frac{t + Kq^2 + \dots}{2} |\tilde{\vec{m}}(\vec{q})|^2.$$

The Gaussian (soluble) theory has independent modes:

$$\langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle_0 = \frac{\delta_{\alpha\beta} \delta_{\vec{q}+\vec{q}',0} V}{t + Kq^2 + Lq^4 + \dots} = \frac{\delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q} + \vec{q}')}{t + Kq^2 + Lq^4 + \dots},$$

using $\sum_{\vec{q}} \delta_{\vec{q},0} = 1 = \int \frac{d^d q}{(2\pi)^d} \delta^d(\vec{q})$. The quartic term is non-local in Fourier space (interactions):

$$\mathcal{U} = u \int \frac{d^d q_1 d^d q_2 d^d q_3}{(2\pi)^{3d}} \tilde{m}_\alpha(\vec{q}_1) \tilde{m}_\alpha(\vec{q}_2) \tilde{m}_\beta(\vec{q}_3) \tilde{m}_\beta(-\vec{q}_1 - \vec{q}_2 - \vec{q}_3) + \dots$$

Wick's theorem: $\langle \prod_{i=1}^l \tilde{m}_{\alpha_i}(\vec{q}_i) \rangle_0$ vanishes if l is odd, and is the sum over all pairings of products of pair averages if l is even.

Averages in perturbation theory.

$$\begin{aligned} \langle \mathcal{O} \rangle &= \frac{\int \mathcal{D}m e^{-(\beta\mathcal{H})_0 - \mathcal{U}} \mathcal{O}}{\int \mathcal{D}m e^{-(\beta\mathcal{H})_0 - \mathcal{U}}} \simeq \frac{\langle \mathcal{O} \rangle_0 - \langle \mathcal{O}\mathcal{U} \rangle_0 + \frac{1}{2} \langle \mathcal{O}\mathcal{U}^2 \rangle_0 + \dots}{1 - \langle \mathcal{U} \rangle_0 + \frac{1}{2} \langle \mathcal{U}^2 \rangle_0 + \dots} \\ &\simeq \langle \mathcal{O} \rangle_0 - (\langle \mathcal{O}\mathcal{U} \rangle_0 - \langle \mathcal{O} \rangle_0 \langle \mathcal{U} \rangle_0) \\ &\quad + \frac{1}{2} (\langle \mathcal{O}\mathcal{U}^2 \rangle_0 - 2 \langle \mathcal{O}\mathcal{U} \rangle_0 \langle \mathcal{U} \rangle_0 - \langle \mathcal{O} \rangle_0 \langle \mathcal{U}^2 \rangle_0 + 2 \langle \mathcal{O} \rangle_0 \langle \mathcal{U} \rangle_0^2) + \dots, \\ &\langle \mathcal{O} \rangle = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \langle \mathcal{O}\mathcal{U}^n \rangle_0^c, \end{aligned}$$

in terms of cumulants (connected averages).

The two-point function. To first order,

$$\begin{aligned} \langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle &= \langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle_0 - u \int \frac{d^d q_1 d^d q_2 d^d q_3}{(2\pi)^{3d}} \Big[\\ &\quad \langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \tilde{m}_i(\vec{q}_1) \tilde{m}_i(\vec{q}_2) \tilde{m}_j(\vec{q}_3) \tilde{m}_j(-\vec{q}_1 - \vec{q}_2 - \vec{q}_3) \rangle_0 \\ &\quad - \langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle_0 \\ &\quad \times \langle \tilde{m}_i(\vec{q}_1) \tilde{m}_i(\vec{q}_2) \tilde{m}_j(\vec{q}_3) \tilde{m}_j(\dots) \rangle_0 \Big] + \dots \end{aligned}$$

Only contractions joining $\mathcal{O}$ with $\mathcal{U}$ survive, of two types:

1. $\tilde{m}_\alpha(\vec{q})$ and $\tilde{m}_\beta(\vec{q}')$ contracted with the same pair $(\tilde{m}_i \tilde{m}_i)$, the other pair contracted with itself: 4 ways, and $\sum_j \delta_{jj} = n$:

$$- \frac{4nu \delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q} + \vec{q}')}{(t + Kq^2 + \dots)^2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots};$$

2. contracted with different pairs: 8 ways,

$$- \frac{8u \delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q} + \vec{q}')}{(t + Kq^2 + \dots)^2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots}.$$

Therefore, with $I(t) := \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots}$,

$$\langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle = \frac{\delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q} + \vec{q}')}{t + Kq^2 + \dots} \left[1 - \frac{4(n+2)u}{t + Kq^2 + \dots} I(t) + \mathcal{O}(u^2) \right].$$

By translational and rotational symmetry,

$$\langle \tilde{m}_\alpha(\vec{q}) \tilde{m}_\beta(\vec{q}') \rangle = \int d^d x d^d x' e^{-i\vec{q}\cdot\vec{x} - i\vec{q}'\cdot\vec{x}'} \langle m_\alpha(\vec{x}) m_\beta(\vec{x}') \rangle = \delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q} + \vec{q}') S(\vec{q}),$$

with the structure factor $S(\vec{q}) = \int d^d r e^{-i\vec{q}\cdot\vec{r}} \langle m_1(\vec{r}) m_1(\vec{0}) \rangle$, Lorentzian $1/(t + Kq^2 + \dots)$ at zeroth order, and $S(\vec{q}=0) = \chi$. Then

$$S^{-1}(\vec{q}) = (t + Kq^2 + \dots) + 4(n+2)u \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots} + \mathcal{O}(u^2),$$

$$\chi^{-1}(t) \simeq t + 4(n+2)u \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots} + \mathcal{O}(u^2).$$

The divergence of χ moves from $t = 0$ to t_c : $\chi^{-1}(t_c) = 0$, and since $t_c \propto u$ it can be neglected inside the integral (it would give $\mathcal{O}(u^2)$ corrections):

$$t_c = -4(n+2)u \int^\Lambda \frac{d^d k}{(2\pi)^d} \frac{1}{Kk^2} = -\frac{4(n+2)u}{K} \frac{S_d}{(2\pi)^d} \frac{\Lambda^{d-2}}{d-2} :$$

the location of the transition changes. Then

$$\chi^{-1}(t) = \chi^{-1}(t) - \chi^{-1}(t_c) = (t - t_c) \left[1 - 4(n+2)u \int \frac{d^d k}{(2\pi)^d} \frac{1}{Kk^2(t - t_c + Kk^2)} + \mathcal{O}(u^2) \right],$$

$$\int^\Lambda \frac{d^d k}{(2\pi)^d} \frac{1}{Kk^2(t - t_c + Kk^2)} \propto \begin{cases} \frac{1}{K^2} \Lambda^{d-4}, & d > 4, \\ \frac{1}{K^2} \left(\frac{t - t_c}{K} \right)^{\frac{d-4}{2}}, & d < 4, \end{cases}$$

$$\chi^{-1}(t) = (t - t_c) \left[1 - A \frac{u(n+2)}{K^2} \cdot \begin{cases} \Lambda^{d-4}, & d > 4 \\ \left(\frac{t-t_c}{K}\right)^{\frac{d-4}{2}}, & d < 4 \end{cases} + \mathcal{O}(u^2) \right] :$$

for $d < 4$ the correction still diverges: perturbation theory diverges. Dimensional analysis of $\int d^d x [tm^2 + K(\nabla m)^2 + um^4]$: $[t][m^2]L^d \sim 1$, $[K][m^2]L^{d-2} \sim 1$, $[u][m^4]L^d \sim 1$, so uL^{4-d}/K^2 is dimensionless, and $\chi = \chi_0[1 + A \frac{u}{K^2} \{a^{4-d}; \xi^{4-d}\}]$: we are really expanding in $u\xi^{4-d}$, not in u . The RG will make sense of this.

4.1 Perturbative RG

1. **Coarse grain:** split $\tilde{m}(\vec{q}) = \tilde{\sigma}(\vec{q})$ for $\Lambda/b < q < \Lambda$ and $\tilde{m}^<(\vec{q})$ for $0 < q < \Lambda/b$ (the singularities are close to the origin):

$$Z = \int \mathcal{D}\tilde{m}^< \mathcal{D}\tilde{\sigma} \exp \left\{ - \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \frac{t + Kq^2 + \dots}{2} |\tilde{m}^<|^2 - \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \frac{t + Kq^2 + \dots}{2} |\tilde{\sigma}|^2 - \mathcal{U}[\tilde{m}^<, \tilde{\sigma}] \right\},$$

where the interaction mixes the two sets. Integrating out $\tilde{\sigma}$,

$$Z = \int \mathcal{D}\tilde{m}^< e^{-\int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \frac{t + Kq^2 + \dots}{2} |\tilde{m}^<|^2} e^{-\frac{nV}{2} \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \ln(t + Kq^2 + \dots)} \left\langle e^{-\mathcal{U}[\tilde{m}^<, \tilde{\sigma}]} \right\rangle_{\sigma},$$

where $\langle \cdot \rangle_{\sigma}$ is the Gaussian average over σ and the second factor is $e^{-V\delta f_b^0}$. Thus $Z = \int \mathcal{D}\tilde{m}^< e^{-\widetilde{\beta\mathcal{H}}[\tilde{m}^<]}$ with

$$\widetilde{\beta\mathcal{H}} = \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \frac{t + Kq^2 + \dots}{2} |\tilde{m}^<|^2 + V\delta f_b^0 - \ln \left\langle e^{-\mathcal{U}[\tilde{m}^<, \tilde{\sigma}]} \right\rangle_{\sigma},$$

and the last term is expanded in cumulants:

$$-\ln \langle e^{-\mathcal{U}} \rangle = \langle \mathcal{U} \rangle - \frac{1}{2} (\langle \mathcal{U}^2 \rangle - \langle \mathcal{U} \rangle^2) + \dots + \frac{(-1)^{l+1}}{l!} \langle \mathcal{U}^l \rangle_c + \dots$$

First order. With $\tilde{m} = \tilde{m}^< + \tilde{\sigma}$,

$$\langle \mathcal{U} \rangle = u \int \frac{d^d q_1 \dots d^d q_4}{(2\pi)^{4d}} (2\pi)^d \delta^d(\vec{q}_1 + \vec{q}_2 + \vec{q}_3 + \vec{q}_4) \left\langle (\tilde{m}_1 + \tilde{\sigma}_1) \cdot (\tilde{m}_2 + \tilde{\sigma}_2) (\tilde{m}_3 + \tilde{\sigma}_3) \cdot (\tilde{m}_4 + \tilde{\sigma}_4) \right\rangle_{\sigma},$$

with $\langle \tilde{\sigma}_{\alpha}(\vec{q}_1) \tilde{\sigma}_{\beta}(\vec{q}_2) \rangle_{\sigma} = \delta_{\alpha\beta} (2\pi)^d \delta^d(\vec{q}_1 + \vec{q}_2) / (t + Kq_1^2)$. The 16 terms are, in diagrammatic form (solid lines: $\tilde{m}^<$; wavy lines: σ):

term	multiplicity	result
[1] $\langle (\tilde{m}_1 \cdot \tilde{m}_2)(\tilde{m}_3 \cdot \tilde{m}_4) \rangle$	1	the original vertex in $\tilde{m}^<$
[2] $\langle (\tilde{\sigma}_1 \cdot \tilde{m}_2)(\tilde{m}_3 \cdot \tilde{m}_4) \rangle$	4	0 (odd in σ)
[3] $\langle (\tilde{\sigma}_1 \cdot \tilde{\sigma}_2)(\tilde{m}_3 \cdot \tilde{m}_4) \rangle$	2	$\propto n$: a closed σ loop on a $\tilde{m}^<$ line
[4] $\langle (\tilde{\sigma}_1 \cdot \tilde{m}_2)(\tilde{\sigma}_3 \cdot \tilde{m}_4) \rangle$	4	a σ line joining the two pairs
[5] $\langle (\tilde{\sigma}_1 \cdot \tilde{\sigma}_2)(\tilde{\sigma}_3 \cdot \tilde{m}_4) \rangle$	4	0 (odd in σ)
[6] $\langle (\tilde{\sigma}_1 \cdot \tilde{\sigma}_2)(\tilde{\sigma}_3 \cdot \tilde{\sigma}_4) \rangle$	1	constant, $uV\delta f_b^1$

(16 terms in total; [6] gives, by Wick's theorem, a “double bubble” plus two exchange diagrams, all independent of $\tilde{m}^<$.)

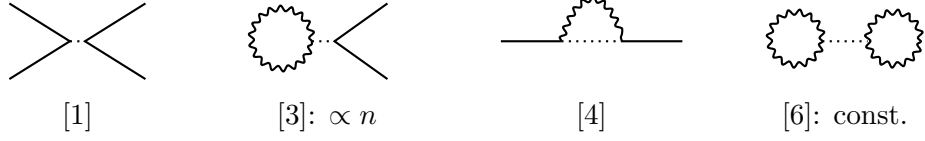


Figure 4.1: First-order diagrams of the coarse-graining step (dotted line: the interaction vertex).

Keeping the non-trivial terms,

$$\begin{aligned}
[3] : & \quad 2u \int \frac{d^d q_1 \cdots d^d q_4}{(2\pi)^{4d}} (2\pi)^d \delta^d(\vec{q}_1 + \cdots + \vec{q}_4) \tilde{m}(\vec{q}_3) \cdot \tilde{m}(\vec{q}_4) \frac{n(2\pi)^d \delta^d(\vec{q}_1 + \vec{q}_2)}{t + Kq_1^2 + \cdots} \\
& = 2un \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} |\tilde{m}(\vec{q})|^2 \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2}, \\
[4] : & \quad 4u \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} |\tilde{m}(\vec{q})|^2 \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2},
\end{aligned}$$

so that, to first order,

$$\widetilde{\beta\mathcal{H}}[\tilde{m}] = V(\delta f_b^0 + u \delta f_b^1) + \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \frac{\tilde{t} + Kq^2 + \cdots}{2} |\tilde{m}(\vec{q})|^2 + u \int_{ba} d^d x m^4 + \mathcal{O}(u^2),$$

with $a \sim 1/\Lambda$ and

$$\tilde{t}_b = t + 4(n+2)u \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \cdots}, \quad \tilde{K}_b = K, \quad \tilde{u}_b = u.$$

Rescale and renormalise. With $\vec{q}' = b\vec{q}$ and $\tilde{m}(\vec{q}^<) = z\tilde{m}'(\vec{q}')$,

$$\begin{aligned}
\beta\mathcal{H}[\tilde{m}'] & = V(\delta f_b^0 + u \delta f_b^1) + \int_0^{\Lambda} \frac{d^d q'}{(2\pi)^d} \frac{b^{-d}(\tilde{t} + Kq'^2 b^{-2} + \cdots) z^2}{2} |\tilde{m}'(\vec{q}')|^2 \\
& \quad + uz^4 b^{-3d} \int \frac{d^d q'_1 d^d q'_2 d^d q'_3}{(2\pi)^{3d}} \tilde{m}'(\vec{q}'_1) \cdot \tilde{m}'(\vec{q}'_2) \tilde{m}'(\vec{q}'_3) \cdot \tilde{m}'(\vec{q}'_4) + \mathcal{O}(u^2),
\end{aligned}$$

so $t' = z^2 b^{-d} \tilde{t}$, $u' = z^4 b^{-3d} u$, $K' = z^2 b^{-d-2} K$. Choosing z such that $K' = K$, $z = b^{1+d/2}$, and

$$t'_b = b^2 \left[t + 4u(n+2) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \cdots} + \mathcal{O}(u^2) \right], \quad u'_b = b^{4-d} u + \mathcal{O}(u^2).$$

RG flows. Choose an infinitesimal rescaling $b = 1 + \delta l$, so that $t'_b = t + \delta l \frac{dt}{dl} + \mathcal{O}(\delta l^2)$, and similarly for u . The shell integral is $\int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} f(k) \simeq K_d \Lambda^{d-1} \delta l f(\Lambda)$, with $K_d := S_d/(2\pi)^d$:

$$t'_b = (1 + 2\delta l) \left[t + 4u(n+2) K_d \frac{\Lambda^d \delta l}{t + K\Lambda^2} \right], \quad u'_b = [1 + (4-d)\delta l] u,$$

$$\frac{dt}{dl} = 2t + \frac{4u(n+2) K_d \Lambda^d}{t + K\Lambda^2} - Au^2, \quad \frac{du}{dl} = (4-d)u - Bu^2,$$

where the u^2 terms come from the next order. The fixed point is $t^* = u^* = 0$ (and, with the second-order term, $u^* = (4 - d)/B$). Linearising around the Gaussian fixed point,

$$\frac{d}{dl} \begin{pmatrix} \delta t \\ \delta u \end{pmatrix} = \begin{pmatrix} 2 & \frac{4(n+2)K_d\Lambda^d}{t + K\Lambda^2} \\ 0 & 4 - d \end{pmatrix} \begin{pmatrix} \delta t \\ \delta u \end{pmatrix},$$

with eigenvalues 2 and $4 - d$.

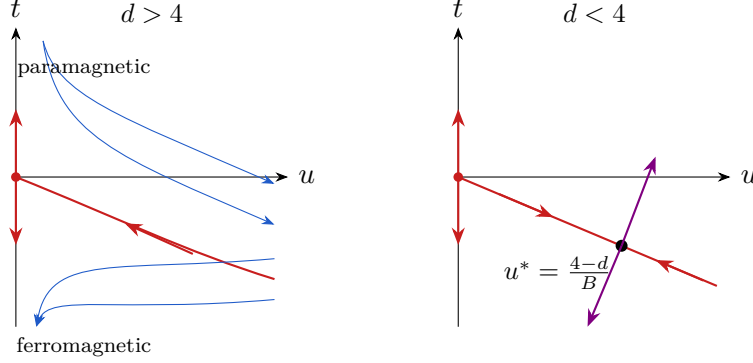


Figure 4.2: RG flows in the (u, t) plane ($u > 0$ for the theory to make sense). For $d > 4$ the Gaussian fixed point controls the transition and the critical surface (red) is its basin of attraction. For $d < 4$ the flow on the critical surface goes to a new fixed point at $u^* = (4 - d)/B$, which requires the next order in u .

For the fixed point to be at small u^* , the perturbation demands to remain close to $d = 4$: in reality the expansion is not in u but in $\varepsilon = 4 - d \ll 1$.

4.2 Second order: the ε expansion

$\mathcal{U} = \int d^d x [um^4 + \dots]$ is necessary for $t < 0$. With $Z = \int \mathcal{D}\tilde{m} e^{-V\delta f_b^0 - (\beta\mathcal{H})_0[\tilde{m}] + \ln\langle e^{-\mathcal{U}[\tilde{m} + \sigma]} \rangle}$,

$$\ln \langle e^{-\mathcal{U}} \rangle = -\langle \mathcal{U} \rangle_0 + \frac{1}{2}(\langle \mathcal{U}^2 \rangle_0 - \langle \mathcal{U} \rangle_0^2) + \dots,$$

and to first order $\langle \mathcal{U} \rangle_0 = \mathcal{U}[\tilde{m}] + u(2n + 4) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots} \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} |\tilde{m}(\vec{q})|^2 + uV\delta f_b^1$.

At second order the connected average $\langle \mathcal{U}^2 \rangle_c$ involves the products of the six types of terms of each $\mathcal{U}$. Products involving term [1] (no σ) are disconnected; several others vanish by parity or because they carry zero momentum. The relevant contributions, which renormalise u , come from products of terms [3] and [4]: the one-loop diagrams with two $\tilde{m}^<$ lines entering and two leaving, with weights $2 \cdot 2 \cdot 2 \cdot n$, $2 \cdot 4 \cdot 2$, $2 \cdot 4 \cdot 2$ and $4 \cdot 4 \cdot 2$. Other products generate an $\tilde{m}^6$ term, corrections to t of order u^2 , and constants ($u^2 V \delta f_b^2$). For instance, the bubble diagram with a closed index loop gives

$$\begin{aligned} & \frac{u^2}{2} \cdot 2 \cdot 2 \cdot 2 \int_0^{\Lambda/b} \frac{d^d q_1 \dots d^d q_4}{(2\pi)^{4d}} \int_{\Lambda/b}^{\Lambda} \frac{d^d k_1 \dots d^d k_2'}{(2\pi)^{4d}} (2\pi)^{2d} \delta^d(\dots) \delta^d(\dots) \\ & \quad \times \tilde{m}(\vec{q}_1) \cdot \tilde{m}(\vec{q}_2) \tilde{m}(\vec{q}_3) \cdot \tilde{m}(\vec{q}_4) \dots \\ & = 4u^2 n \int_0^{\Lambda/b} \frac{d^d q_1 \dots d^d q_4}{(2\pi)^{4d}} \tilde{m}(\vec{q}_1) \cdot \tilde{m}(\vec{q}_2) \tilde{m}(\vec{q}_3) \cdot \tilde{m}(\vec{q}_4) \\ & \quad \times (2\pi)^d \delta^d(\vec{q}_1 + \dots + \vec{q}_4) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{(t + Kk^2)(t + K(\vec{q}_1 + \vec{q}_2 + \vec{k})^2)}, \end{aligned}$$

and expanding the last integral, $\int \frac{d^d k}{(2\pi)^d} \frac{1}{(t + Kk^2)^2} [1 + \frac{K(\dots)^2}{t + Kk^2}]^{-1}$, generates momentum-dependent terms such as $m^2(\nabla m)^2$: non-linearities that did not appear at the beginning, which is why they should be included from the start. Collecting everything,

$$\begin{aligned} \widetilde{\beta\mathcal{H}} = & V \left(\delta f_b^0 + u \delta f_b^1 + \frac{u^2}{2} \delta f_b^2 \right) + \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} |\tilde{m}(\vec{q})|^2 \frac{\tilde{t} + Kq^2 + \dots}{2} \\ & + \int_0^{\Lambda/b} \frac{d^d q_1 \dots d^d q_4}{(2\pi)^{4d}} (2\pi)^d \delta^d(\vec{q}_1 + \dots + \vec{q}_4) \tilde{m}_1 \cdot \tilde{m}_2 \tilde{m}_3 \cdot \tilde{m}_4 \\ & \times \left[u - \frac{u^2}{2} (8n + 64) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{(t + Kk^2 + \dots)^2} + \mathcal{O}(q^2) \right], \end{aligned}$$

with $\tilde{t} = t + 4u(n + 2) \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2} - u^2 A + \dots$. The recursion relations become

$$\frac{dt}{dl} = 2t + \frac{4u(n + 2)K_d \Lambda^d}{t + K\Lambda^2 + \dots} - A(t, K, \dots)u^2, \quad \frac{du}{dl} = \varepsilon u - \frac{4(n + 8)u^2 K_d \Lambda^d}{(t + K\Lambda^2 + \dots)^2}, \quad \varepsilon = 4 - d.$$

Fixed points. The Gaussian one, $t^* = u^* = 0$, and the $O(n)$ (Wilson–Fisher) fixed point,

$$u^* = \frac{(t^* + K\Lambda^2 + \dots)^2}{4(n + 8)K_d \Lambda^d} \varepsilon, \quad t^* = -\frac{2(n + 2)K_d \Lambda^d}{t^* + K\Lambda^2 + \dots} u^*,$$

and, since $t^* \sim \varepsilon$,

$$u^* = \frac{K^2 \Lambda^{4-d}}{4(n + 8)K_d} \varepsilon + \mathcal{O}(\varepsilon^2), \quad t^* = -\frac{(n + 2)}{2(n + 8)} K \Lambda^2 \varepsilon + \mathcal{O}(\varepsilon^2).$$

Linearising with $t = t^* + \delta t$, $u = u^* + \delta u$,

$$\frac{d}{dl} \begin{pmatrix} \delta t \\ \delta u \end{pmatrix} = \begin{pmatrix} 2 - \frac{4u^*(n + 2)K_d \Lambda^d}{(t^* + K\Lambda^2)^2} & \frac{4(n + 2)K_d \Lambda^d}{t^* + K\Lambda^2} - 2Au^* \\ \mathcal{O}(\varepsilon^2) & \varepsilon - \frac{8(n + 8)K_d \Lambda^d u^*}{(t^* + K\Lambda^2)^2} \end{pmatrix} \begin{pmatrix} \delta t \\ \delta u \end{pmatrix},$$

so that

$$y_u = \varepsilon - 2\varepsilon = -\varepsilon, \quad y_t = 2 - \frac{n + 2}{n + 8} \varepsilon, \quad \nu = \frac{1}{y_t} = \frac{1}{2} + \frac{1}{4} \frac{n + 2}{n + 8} \varepsilon + \mathcal{O}(\varepsilon^2).$$

For $d < 4$ the Gaussian fixed point has two relevant directions; the $O(n)$ fixed point has a single one (u is irrelevant there): at 3.999 dimensions there is only one relevant direction besides h .

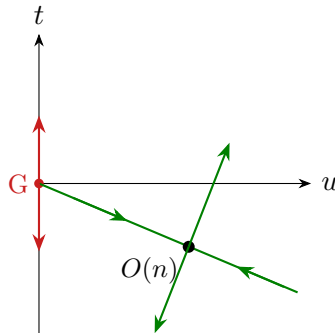


Figure 4.3: For $d < 4$ the critical behaviour is controlled by the $O(n)$ fixed point.

5 Perturbative renormalisation group: summary and results

The critical exponents are material independent (collective behaviour). At first order in ε , with $\varepsilon = 1$:

	α	β	γ	ν
ferromagnets	-0.12 (0.05)	0.37	1.39	0.71 (0.61)
superfluids	-0.01 (0.10)	0.35	1.32	0.67 (0.60)
liquid-gas	0.11 (0.17)	0.33	1.24	0.63 (0.58)

(in brackets: $\mathcal{O}(\varepsilon)$ predictions at $\varepsilon = 1$).

The steps of the theory:

1. **Landau:** a statistical field $\vec{m}(\vec{x} \in \mathbb{R}^d) \in \mathbb{R}^n$ with $Z = \int \mathcal{D}\vec{m} e^{-\beta\mathcal{H}[\vec{m}]}$,

$$\beta\mathcal{H} = \int d^d x \left[\underbrace{\frac{t}{2}m^2 + \frac{K}{2}(\nabla m)^2 + \frac{L}{2}(\nabla^2 m)^2 + \dots}_{(\beta\mathcal{H})_0} + \underbrace{um^4 + \dots + vm^2(\nabla m)^2 + u_6 m^6 + \dots - \vec{h} \cdot \vec{m}}_{\mathcal{U}} \right],$$

with all the terms consistent with locality, symmetry and analyticity ($\vec{\mu}(T) = (t, K, u, v, \dots)$ analytic, since a finite number of degrees of freedom is coarse-grained). The saddle-point approximation and naive perturbation theory fail.

2. **Scaling** (Widom, Fisher, ...): $f_{\text{sing}}(t, h) = t^{2-\alpha} g(h/t^\Delta)$ gives the exponent identities; self-similarity.
3. **Kadanoff RG:** $\vec{\mu}'_b = \vec{\mu}'_b(\vec{\mu})$ has fixed points $\vec{\mu}^*$; $\delta\vec{\mu}'_b = \mathcal{L}_b \delta\vec{\mu}_b$ gives the relevant variables.
4. **Wilson: perturbative RG:** (1) coarse grain, $\widetilde{\beta\mathcal{H}} = V\delta f_b^0 + (\beta\mathcal{H})_0[\vec{m}] - \ln \langle e^{-\mathcal{U}[\vec{m}, \sigma]} \rangle_\sigma$, with $-\ln \langle e^{-\mathcal{U}} \rangle = \langle \mathcal{U} \rangle_0 - \frac{1}{2}(\langle \mathcal{U}^2 \rangle_0 - \langle \mathcal{U} \rangle_0^2) + \dots$; (2) rescale, $\vec{q} \rightarrow b^{-1}\vec{q}'$; (3) renormalise, $\vec{m} \rightarrow z\vec{m}'$.

The general recursion relations are

$$\begin{aligned} t' &= b^{-d} z^2 \left[t + 4u(n+2) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + Kk^2 + \dots} \right. \\ &\quad \left. - A_1(t, K, \Lambda) u^2 + \mathcal{O}(v^2, uv, u_6, \dots) \right], \\ K' &= b^{-d-2} z^2 \left[K - A_2(t, K, \Lambda) u^2 + \mathcal{O}(v^2, uv, u_6, \dots) \right], \\ L' &= b^{-d-4} z^2 \left[L - A_3(t, K, \Lambda) u^2 + \mathcal{O}(v^2, uv, u_6, \dots) \right], \\ u' &= b^{-3d} z^4 \left[u - 4u^2(n+8) \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{(t + Kk^2 + \dots)^2} \right. \\ &\quad \left. + \mathcal{O}(uv, u_6, u^3, \dots) \right], \\ v' &= b^{-3d-2} z^4 \left[v - \mathcal{O}(uv, u_6, \dots) \right], \quad u'_6 = b^{-5d} z^6 \left[u_6 - \mathcal{O}(u^2 v, \dots) \right], \end{aligned}$$

where K only gets a contribution at order u^2 (the momentum gets into the loops through the two-loop “sunset” diagram). Choosing z such that $K' = K$ (this is where the physics enters):

$K'/K = z^2 b^{-d-2} [1 + \mathcal{O}(u^2)]$, so $z = b^{1+d/2} + \mathcal{O}(\varepsilon^2)$. With $b = 1 + \delta l$,

$$\begin{aligned}\frac{dt}{dl} &= 2t + \frac{4u(n+2)K_d\Lambda^d}{t + K\Lambda^2 + \dots} - A_1 u^2 + \mathcal{O}(u^3, uv, u_6, \dots), \\ \frac{du}{dl} &= \varepsilon u - \frac{4u^2(n+8)K_d\Lambda^d}{(t + K\Lambda^2 + \dots)^2} + \mathcal{O}(u^3, uv, u_6, \dots) \quad (\text{Fisher}), \\ \frac{dL}{dl} &= -2L + \mathcal{O}(u^2, uv, \dots), \quad \frac{dv}{dl} = (-2 + \varepsilon)v + \mathcal{O}(u^2, uv, \dots), \\ \frac{du_6}{dl} &= (-2 + 2\varepsilon)u_6 + \mathcal{O}(uv, \dots).\end{aligned}$$

At the $O(n)$ fixed point, $L^* \sim v^* \sim \mathcal{O}(\varepsilon^2)$ and $u_6^* \sim \mathcal{O}(\varepsilon^3)$. The eigenvalues (for the Gaussian fixed point, the “bare dimensions”; $h' = zh = b^{1+d/2+\mathcal{O}(\varepsilon)}h$) are:

	t	u	L	v	u_6	h
Gaussian	2	ε	-2	$-2 + \varepsilon$	$-2 + 2\varepsilon$	$1 + \frac{d}{2}$
$O(n)$	$2 - \frac{n+2}{n+8}\varepsilon$	$-\varepsilon$	$-2 + \mathcal{O}(\varepsilon)$	$-2 + \mathcal{O}(\varepsilon)$	$-2 + \mathcal{O}(\varepsilon)$	$1 + \frac{d}{2} + \mathcal{O}(\varepsilon)$

Then

$$\begin{aligned}\nu &= \frac{1}{y_t} = \frac{1}{2} \left[1 - \frac{n+2}{2(n+8)}\varepsilon \right]^{-1} = \frac{1}{2} + \frac{1}{4} \frac{n+2}{n+8} \varepsilon, \\ \alpha &= 2 - d\nu = 2 - (4 - \varepsilon) \frac{1}{2} \left[1 + \frac{1}{2} \frac{n+2}{n+8} \varepsilon + \dots \right] = \frac{4-n}{2(n+8)} \varepsilon.\end{aligned}$$

How do we know there is not another fixed point? Perturbatively there is no proof, but neither experiments nor simulations show the necessity or existence of such a point.

Higher orders. For $n = 1$,

$$\gamma = 1 + 0.167\varepsilon + 0.077\varepsilon^2 - 0.049\varepsilon^3 + 0.180\varepsilon^4 - 0.415\varepsilon^5 + \dots,$$

whose partial sums at $\varepsilon = 1$ are 1, 1.167, 1.244, 1.195, 1.375, 0.96: an **asymptotic series**, with coefficients growing as $|f_p| \sim c p! a^p$. **Borel summation**, using $p! = \int_0^\infty dx x^p e^{-x}$,

$$\sum_p f_p \varepsilon^p = \sum_p f_p \varepsilon^p \int_0^\infty dx \frac{x^p e^{-x}}{p!} = \int_0^\infty dx e^{-x} \underbrace{\sum_p \frac{f_p(\varepsilon x)^p}{p!}}_{B(\varepsilon x)},$$

gives $\gamma = 1.2385 \pm 0.0025$.

The limit $\varepsilon \ll 1$ has an analogue in $n \gg 1$: the spherical model and the $1/n$ expansion. In the (n, d) plane, mean-field behaviour holds for $d > 4$, the ε expansion works down from $d = 4$, the $1/n$ expansion inward from large n , and $d = 2$ is the lower critical dimension for continuous symmetries.

6 Position-space renormalisation group

6.1 Lattice models

On a lattice with spins $\vec{s}_i$ at the sites,

$$Z = \sum_{\{\vec{s}_i\}} e^{-\sum_{\langle ij \rangle} \beta \hat{B}(\vec{s}_i, \vec{s}_j)},$$

with the sum over nearest neighbours and $\hat{B}$ the energy per bond. Some simple models:

- **Ising model:** binary variables $\sigma_i = \pm 1$; with z the coordination number, $\hat{B}(\sigma_i, \sigma_j) = -\hat{g} - \frac{\hat{h}}{z}(\sigma_i + \sigma_j) - \hat{J}\sigma_i\sigma_j$ (since $\sigma_i^2 = 1$ the series terminates here), so

$$Z = \sum_{\{\sigma_i\}} \exp \left[g + h \sum_i \sigma_i + K \sum_{\langle ij \rangle} \sigma_i \sigma_j \right],$$

a sum over 2^N states.

- **$O(n)$ model:** $\vec{s}_i \in \mathbb{R}^n$, $|\vec{s}_i| = 1$ ($n = 1$: Ising; $n = 2$: XY; $n = 3$: Heisenberg).
- **Potts model:** $s_i \in \{1, 2, \dots, q\}$, $-\beta\mathcal{H} = \sum_{\langle ij \rangle} K \delta_{s_i, s_j}$, with permutation symmetry ($q = 2$ is the Ising model).
- **Spin- s model:** $s_i \in \{-s, -s+1, \dots, s\}$; classically equivalent to the Ising model (same universality class).

Tools to solve lattice models: (1) position-space RG; (2) exact solutions; (3) series expansions (high and low temperature); (4) Monte Carlo simulations.

6.2 Position-space RG of the one-dimensional Ising model

For a chain of N spins $\sigma_i = \pm 1$,

$$Z = \sum_{\{\sigma_i\}} e^{\sum_i B(\sigma_i, \sigma_{i+1})}, \quad B(\sigma_i, \sigma_{i+1}) = g + \frac{h}{2}(\sigma_i + \sigma_{i+1}) + K\sigma_i\sigma_{i+1}.$$

RG with $b = 2$. A majority rule $\sigma'_i = (\sigma_i + \sigma_{i+1})/2 \in \{+1, -1, 0\}$ with a tie breaker (taking the left spin when 0) is equivalent to removing every other spin (**decimation**): rename $\{\sigma_i\} \rightarrow \{\sigma'_i, s_i\}$, $i = 1, \dots, N/2$, with $\sigma'_i = \sigma_{2i-1}$, $s_i = \sigma_{2i}$. Then

$$Z = \sum_{\{\sigma'_i\}} \sum_{\{s_i\}} e^{\sum_{i=1}^{N/2} [B(\sigma'_i, s_i) + B(s_i, \sigma'_{i+1})]} = \sum_{\{\sigma'_i\}} \prod_{i=1}^{N/2} \underbrace{\left(\sum_{s_i = \pm 1} e^{B(\sigma'_i, s_i) + B(s_i, \sigma'_{i+1})} \right)}_{e^{B'(\sigma'_i, \sigma'_{i+1})}} = \sum_{\{\sigma'_i\}} e^{\sum_i B'(\sigma'_i, \sigma'_{i+1})},$$

with new parameters g' , h' , K' :

$$e^{g' + \frac{h'}{2}(\sigma'_i + \sigma'_{i+1}) + K'\sigma'_i\sigma'_{i+1}} = \sum_{s_i = \pm 1} e^{2g + \frac{h}{2}(\sigma'_i + s_i) + K\sigma'_i s_i + \frac{h}{2}(s_i + \sigma'_{i+1}) + K s_i \sigma'_{i+1}}.$$

With $x = e^K$, $y = e^h$, $z = e^g$ (and primes), the four configurations of $(\sigma'_i, \sigma'_{i+1})$ give

$$\begin{aligned} (+, +) : \quad z' y' x' &= z^2 y (x^2 y + x^{-2} y^{-1}), & (-, -) : \quad z' y'^{-1} x' &= z^2 y^{-1} (x^{-2} y + x^2 y^{-1}), \\ (+, -) : \quad z' x'^{-1} &= z^2 (y + y^{-1}), & (-, +) : \quad z' x'^{-1} &= z^2 (y + y^{-1}), \end{aligned}$$

so that

$$\begin{aligned} z'^4 &= z^8 (x^2 y + x^{-2} y^{-1}) (x^{-2} y + x^2 y^{-1}) (y + y^{-1})^2, \\ y'^2 &= y^2 \frac{x^2 y + x^{-2} y^{-1}}{x^{-2} y + x^2 y^{-1}}, \quad x'^4 = \frac{(x^2 y + x^{-2} y^{-1}) (x^{-2} y + x^2 y^{-1})}{(y + y^{-1})^2}, \end{aligned}$$

i.e. $g' = 2g + \delta g(K, h)$, $h' = h + \delta h(K, h)$, $K' = K'(K, h)$.

Zero field. For $h = 0$, $y = 1 = y'$, so $h' = 0$, and $x'^4 = (x^2 + x^{-2})^2/4$:

$$K' = \frac{1}{2} \ln \cosh(2K) \quad \Longleftrightarrow \quad \tanh K' = \tanh^2 K.$$

Fixed points: for $K \rightarrow 0$, $K' = \frac{1}{2} \ln[1 + (2K)^2/2 + \dots] \simeq K^2$; for $K \rightarrow \infty$, $e^{4K'} \simeq e^{4K}/4$, i.e. $K' = K - \frac{1}{2} \ln 2$. All finite K flow to $K = 0$: the 1D system will not order. Near $K \rightarrow \infty$ ($\tanh K = 1$), $y'^2 = y^2 \cdot y/y^{-1} = y^4$, so $h' = 2h = b^{y_h} h$ with $y_h = 1$ (a relevant direction), and $e^{-K'} = \sqrt{2} e^{-K}$.

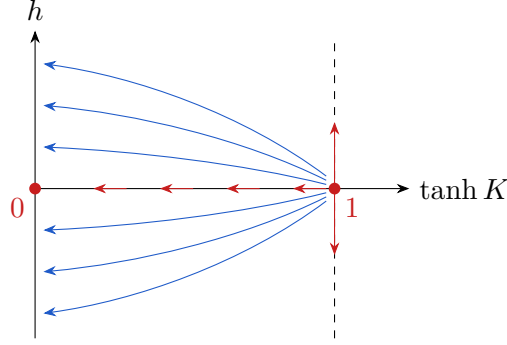


Figure 6.1: RG flow of the 1D Ising model: the zero-temperature fixed point ($\tanh K = 1$) is unstable, with h relevant.

Correlation length. Before and after one RG step, $\xi(h, e^{-K}) = 2 \xi(2h, \sqrt{2} e^{-K})$: all length scales have shrunk by a factor 2. After l steps, $\xi(h, e^{-K}) = 2^l \xi(2^l h, 2^{l/2} e^{-K})$; choosing $2^{l/2} e^{-K} \sim 1$, i.e. $2^l \sim e^{2K}$,

$$\xi = e^{2K} g(h e^{2K}), \quad f_{\text{sing}} \sim \xi^{-d} = e^{-2K} g_2(h e^{2K}),$$

$$\chi \sim \frac{\partial^2 f}{\partial h^2} \Big|_{h=0} \sim e^{2K} \sim \int d^d x \langle \sigma_0 \sigma_x \rangle \sim \int d^d x \frac{e^{-x/\xi}}{|x|^{d-2+\eta}} \sim \xi^{2-\eta},$$

so $\eta = 1$ (with $\langle \sigma_0 \sigma_x \rangle \sim e^{-x/\xi}$). This model has no phase transition, but shows its signatures, like the divergence of the correlation length and susceptibility at strong (nearest-neighbour) coupling.

Decimation in 2D. On the square lattice ($h = 0$), summing over a spin s with neighbours $\sigma_1, \dots, \sigma_4$,

$$\sum_s e^{Ks(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4)} = 2 \cosh[K(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4)]$$

$$= e^{g' + K'(\sigma_1 \sigma_2 + \sigma_2 \sigma_3 + \sigma_3 \sigma_4 + \sigma_4 \sigma_1 + \sigma_1 \sigma_3 + \sigma_2 \sigma_4) + K_4 \sigma_1 \sigma_2 \sigma_3 \sigma_4} ;$$

decimation generates next-nearest-neighbour and four-spin interactions. Eliminating whole rows instead still generates long-range interactions.

6.3 Transfer matrices

With periodic boundary conditions,

$$Z = \sum_{\{s_i\}} \prod_i e^{B(s_i, s_{i+1})} = \sum_{s_1, s_2, \dots} \langle s_1 | T | s_2 \rangle \langle s_2 | T | s_3 \rangle \cdots \langle s_N | T | s_1 \rangle = \text{Tr } T^N = \sum_{\alpha} \lambda_{\alpha}^N \simeq \lambda_{\text{max}}^N,$$

with $\langle s|T|s' \rangle = e^{B(s,s')}$. **Frobenius (Perron–Frobenius) theorem:** if all the elements of a matrix are positive, its largest eigenvalue is non-degenerate. Hence $\lambda_{\max}$ is an analytic function of the parameters of the Hamiltonian (there is no crossing of eigenvalues): a mathematical way of saying that there is no phase transition in 1D. A 2D lattice of size $L \times L$ can be seen as a complicated 1D system with a $2^L \times 2^L$ transfer matrix. By the Frobenius theorem there should be no phase transition, but Lars Onsager proved otherwise by diagonalising this matrix: for $L \rightarrow \infty$ the theorem no longer applies.

6.4 Cumulant expansion (Niemeijer and van Leeuwen)

For the Ising model $Z = \sum_{\{\sigma_i\}} e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + h \sum_i \sigma_i}$, decimation works in 1D ($b = 2$: $K' = \frac{1}{2} \ln \cosh 2K$); in 2D one can decimate a sublattice ($b = \sqrt{2}$). On the triangular lattice, group the sites into triangular cells α , β and assign a cell spin by the majority rule, $\sigma'_\alpha = \text{maj}(\sigma_\alpha^1, \sigma_\alpha^2, \sigma_\alpha^3) \in \{\pm 1\}$ ($b = \sqrt{3}$). Split

$$\begin{aligned} -\beta \mathcal{H} &= K \sum_{\langle ij \rangle} \sigma_i \sigma_j = -(\beta \mathcal{H})_0 - \mathcal{U}, \\ -(\beta \mathcal{H})_0 &= \sum_{\text{cells } \alpha} K(\sigma_1^\alpha \sigma_2^\alpha + \sigma_2^\alpha \sigma_3^\alpha + \sigma_3^\alpha \sigma_1^\alpha), \\ -\mathcal{U} &= \sum_{\langle \alpha \beta \rangle} K(\sigma_1^\beta \sigma_2^\alpha + \sigma_1^\beta \sigma_3^\alpha), \end{aligned}$$

the intra- and inter-cell couplings. With the sum restricted to configurations compatible with $\{\sigma'_\alpha\}$,

$$\begin{aligned} W[\sigma'_\alpha] &= \sum_{\{\sigma_i\}|\sigma'} e^{-(\beta \mathcal{H})_0} \left[1 - \mathcal{U} + \frac{\mathcal{U}^2}{2} - \dots \right] = Z_0[\sigma'] \left[1 - \langle \mathcal{U} \rangle_0 + \frac{1}{2} \langle \mathcal{U}^2 \rangle_0 - \dots \right], \\ -(\beta \mathcal{H})'[\sigma'_\alpha] &= \ln W[\sigma'_\alpha] = \ln Z_0[\sigma'] - \langle \mathcal{U} \rangle_0 + \frac{1}{2} (\langle \mathcal{U}^2 \rangle_0 - \langle \mathcal{U} \rangle_0^2) + \dots \end{aligned}$$

For each value of σ'_α , four internal configurations are compatible (all three spins aligned, weight e^{3K} , or one flipped, weight e^{-K}), so $\ln Z_0 = \frac{N}{3} \ln(e^{3K} + 3e^{-K})$ and

$$\langle \sigma_\alpha^i \rangle_0 = \frac{e^{3K} + e^{-K}}{e^{3K} + 3e^{-K}} \sigma'_\alpha.$$

Two spins of each cell couple to a neighbouring cell, and within $\langle \cdot \rangle_0$ different cells are independent:

$$\begin{aligned} -\langle \mathcal{U} \rangle_0 &= K \sum_{\langle \alpha \beta \rangle} (\langle \sigma_1^\beta \rangle_0 \langle \sigma_2^\alpha \rangle_0 + \langle \sigma_1^\beta \rangle_0 \langle \sigma_3^\alpha \rangle_0) = 2K \sum_{\langle \alpha \beta \rangle} \left(\frac{e^{3K} + e^{-K}}{e^{3K} + 3e^{-K}} \right)^2 \sigma'_\alpha \sigma'_\beta, \\ K' &= 2K \left(\frac{e^{3K} + e^{-K}}{e^{3K} + 3e^{-K}} \right)^2. \end{aligned}$$

For $K \rightarrow 0$, $K' = K/2$; for $K \rightarrow \infty$, $K' = 2K$; the non-trivial fixed point satisfies $\sqrt{2}(e^{3K^*} + e^{-K^*}) = e^{3K^*} + 3e^{-K^*}$, i.e.

$$K^* = \frac{1}{4} \ln \frac{3 - \sqrt{2}}{\sqrt{2} - 1} \simeq 0.34,$$

to be compared with the exact $K_c = \frac{1}{4} \ln 3 \simeq 0.27$. Then $\partial K' / \partial K|_{K^*} \simeq 1.62 = b^{y_t} = (\sqrt{3})^{y_t}$, so $y_t \simeq 0.88$ (exact: $y_t = 1$), and $\nu = 1/y_t \simeq 1.13$, $\alpha = 2 - 2\nu \simeq -0.26$. At $\mathcal{O}(\mathcal{U}^2)$ next-nearest and further neighbour couplings K' , L' , M' are generated, and $y_t \simeq 1.05$.

To obtain y_h , add h as an additional perturbation: $\langle h \sum_\alpha (\sigma_\alpha^1 + \sigma_\alpha^2 + \sigma_\alpha^3) \rangle_0 = 3h \sum_\alpha \frac{e^{3K} + e^{-K}}{e^{3K} + 3e^{-K}} \sigma'_\alpha$, so $h' = 3h \frac{e^{3K} + e^{-K}}{e^{3K} + 3e^{-K}}$ and $b^{y_h} = \partial h' / \partial h|_{K^*} = 3/\sqrt{2}$: $y_h \simeq 1.4$ (exact: 1.875).

6.5 Migdal–Kadanoff bond moving

On the square lattice with $b = 2$, move the bonds of every other row/column onto their neighbours (doubling their strength) so that the intermediate spins only have two bonds, and then decimate as in 1D:

$$K' = \frac{1}{2} \ln \cosh(4K).$$

For $K \rightarrow 0$, $K' = \frac{1}{2} \ln[1 + (4K)^2/2 + \dots] \simeq 4K^2$; for $K \rightarrow \infty$, $K' \simeq 2K$; the fixed point $K^* = \frac{1}{2} \ln \cosh(4K^*)$ gives $K^* \simeq 0.30$ (exact: $K_c \simeq 0.44$), and $b^{y_t} = 2^{y_t} = \partial K'/\partial K|_{K^*}$ gives $y_t \simeq 0.75$ (exact: 1). For a d -dimensional hypercubic lattice,

$$K' = \frac{1}{2} \ln \cosh(2^{d-1} \cdot 2K), \quad K'_b = \frac{1}{2} \ln \cosh(b^{d-1} \cdot 2K), \quad K'_b \xrightarrow{K \rightarrow \infty} b^{d-1} K.$$

General spin model. For $Z = \sum_{\{s_i\}} \prod_i e^{K(s_i, s_{i+1})} = \text{Tr } T^N$ with $T = e^{K(s, s')}$, $Z = \text{Tr}(T^b)^{N/b}$, so in 1D $T'_b = T^b$, i.e. $T'_b(\{K'\}) = T(\{K\})^b$. The Migdal–Kadanoff RG is

$$T'_b(\{K'\}) = T(\{b^{d-1} K\})^b.$$

In 1D, $Z = \text{Tr } T^N = \sum_\alpha \lambda_\alpha^N \rightarrow \lambda_{\max}^N$, with $e^{K(s, s')}$ a matrix of positive elements (for Ising, $\begin{pmatrix} e^K & e^{-K} \\ e^{-K} & e^K \end{pmatrix}$). Non-analyticity requires two eigenvalues to cross, which does not happen if all elements are positive (analogy in quantum mechanics: the ground state in a potential is not degenerate). An $L \times N$ strip with $N \rightarrow \infty$ has no phase transition (it is an effective 1D system once the correlated regions reach the width L); an $N \times N$ lattice with $N \rightarrow \infty$ does.

7 Series expansions

By universality, the liquid–gas critical point ($n = 1$) is described by a binary model: the Ising model, $\sigma_i = \pm 1$, with $-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \sigma_i \sigma_j$, $K = J/k_B T$, and $Z(K) = \sum_{\{\sigma_i\}} e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j}$.

7.1 Low-temperature expansion

At $T = 0$ all $\sigma_i = +1$ or all $\sigma_i = -1$. Excitations are islands of $-$ spins in a sea of $+$, each broken bond costing e^{-2K} :

$$Z_{T \rightarrow 0} = 2e^{dNK} \left[1 + \sum_{\text{islands}} e^{-2K(\# \text{broken bonds})} \right].$$

excitation	degeneracy	broken bonds
one flipped spin	N	$2d$
two adjacent flipped spins	dN	$2(2d - 1)$
two separated flipped spins	$N(N - 2d - 1)/2$	$4d$

$$Z = 2e^{dNK} \left[1 + Ne^{-4dK} + dNe^{-4(2d-1)K} + \frac{N(N - 2d - 1)}{2} e^{-8dK} + \dots \right].$$

With $\ln(1 + x) = x - x^2/2 + \dots$, the subtraction of $\frac{N^2}{2} e^{-8dK}$ ensures that the result is of order N (disconnected islands cancel):

$$\begin{aligned} \frac{E}{N} &= -\frac{J}{N} \frac{\partial \ln Z}{\partial K} = -J \frac{\partial}{\partial K} \left[\frac{\ln 2}{N} + dK + e^{-4dK} + de^{-4(2d-1)K} - \frac{2d+1}{2} e^{-8dK} + \dots \right] \\ &= -J[d - 4de^{-4dK} - 4d(2d-1)e^{-4(2d-1)K} + 4d(2d+1)e^{-8dK} + \dots], \end{aligned}$$

$$\begin{aligned}\frac{C}{k_B N} &= \frac{d(E/N)}{d(k_B T)} = -\frac{K^2}{J} \frac{\partial}{\partial K} \left(\frac{E}{N} \right) \\ &= K^2 [16d^2 e^{-4dK} + 16d(2d-1)^2 e^{-4(2d-1)K} - 32d^2(2d+1)e^{-8dK} + \dots].\end{aligned}$$

Can the exponent and the location of the singularity be found from the series? With $u = e^{-K}$ (or a similar variable), an “expert” assumes that as $u \rightarrow u_c$, $C = A(1 - u/u_c)^{-\alpha}$, i.e.

$$C = A \left[1 + \alpha \frac{u}{u_c} + \frac{\alpha(\alpha+1)}{2} \frac{u^2}{u_c^2} + \dots + \frac{\alpha(\alpha+1) \dots (\alpha+l-1)}{l!} \frac{u^l}{u_c^l} + \dots \right] = \sum_l a_l u^l.$$

Ratio method.

$$\frac{a_l}{a_{l-1}} = \frac{\alpha + l - 1}{l u_c} = \frac{1}{u_c} \left[1 - \frac{1 - \alpha}{l} \right] :$$

plotted against $1/l$, the ratios lie on a line with intercept u_c^{-1} and slope $-(1 - \alpha)/u_c$. This gives $\alpha \simeq 0$ in $d = 2$ and $\alpha \simeq 0.11$ in $d = 3$. It could fail if $C = A(1 - u/u_c)^{-\alpha} + \sum_l b_l u^l$ with large regular contributions, or if the coefficients alternate in sign (here the next coefficient is negative): e.g. $\frac{1}{1-z/2} \frac{1}{1+z^2} = (1 + \frac{z}{2} + \frac{z^2}{4} + \dots)(1 - z^2 + z^4 - \dots)$ has poles at $z = \pm i$ that limit the radius of convergence before the physical singularity at $z = 2$. A change of variables such as $v = \tanh K = (e^{2K} - 1)/(e^{2K} + 1)$ and **Padé approximants** help.

7.2 High-temperature expansion

At $T \rightarrow \infty$ the spins take ± 1 with equal probability. Then

$$\begin{aligned}Z &= \text{Tr } e^{-\beta \mathcal{H}} = \text{Tr} \left[1 - \beta \mathcal{H} + \frac{(\beta \mathcal{H})^2}{2} - \dots \right] = Z_0 \left[1 - \langle \beta \mathcal{H} \rangle_0 + \frac{1}{2} \langle (\beta \mathcal{H})^2 \rangle_0 - \dots \right], \\ \ln Z &= \ln Z_0 + \sum_{l=1}^{\infty} \frac{(-1)^l}{l!} \langle (\beta \mathcal{H})^l \rangle_{0,c},\end{aligned}$$

with $Z_0 = 2^N$. For $-\beta \mathcal{H} = K \sum_{\langle ij \rangle} \sigma_i \sigma_j$: $-\langle \beta \mathcal{H} \rangle_0 = K \sum \langle \sigma_i \rangle_0 \langle \sigma_j \rangle_0 = 0$, and $\langle (\beta \mathcal{H})^2 \rangle_0 = K^2 \sum_{\langle ij \rangle, \langle kl \rangle} \langle \sigma_i \sigma_j \sigma_k \sigma_l \rangle_0 = K^2 \sum_{\langle ij \rangle} 1 = K^2 dN$ (the number of bonds). A more efficient organization: for each bond,

$$e^{K \sigma_i \sigma_j} = \frac{e^K + e^{-K}}{2} + \sigma_i \sigma_j \frac{e^K - e^{-K}}{2} = \cosh K (1 + t \sigma_i \sigma_j), \quad t = \tanh K,$$

$$Z = \sum_{\{\sigma_i\}} \prod_{\langle ij \rangle} \cosh K (1 + t \sigma_i \sigma_j) = (\cosh K)^{\# \text{bonds}} \sum_{\{\sigma_i\}} \prod_{\langle ij \rangle} (1 + t \sigma_i \sigma_j),$$

$$Z = 2^N (\cosh K)^{\# \text{bonds}} \sum_{\substack{\text{graphs with an even} \\ \text{number of bonds per site}}} t^{\# \text{bonds in graph}},$$

since the sum over σ_i vanishes for any site touched by an odd number of bonds. On the hypercubic lattice,

$$Z = 2^N (\cosh K)^{dN} \left[1 + \frac{d(d-1)}{2} N t^4 + Nd(d-1)(2d-3)t^6 + \dots \right],$$

from the elementary plaquettes and the six-bond loops. High-temperature series can be used to (1) solve exactly the $d = 1$ model, (2) re-derive the Gaussian model, (3) solve exactly the 2D Ising model.

One-dimensional Ising model. (a) Open chain of N sites: no closed graphs exist, so $Z = 2^N (\cosh K)^{N-1} \times 1$. For the correlation function,

$$\langle \sigma_m \sigma_n \rangle = \frac{\sum_{\{\sigma_i\}} \prod_{\langle ij \rangle} e^{K \sigma_i \sigma_j} \sigma_m \sigma_n}{Z} = \frac{2^N (\cosh K)^{\# \text{bonds}} \sum_g t^{\# \text{bonds in } g}}{2^N (\cosh K)^{\# \text{bonds}} \sum_{\text{all graphs}} t^{\# \text{bonds}}},$$

where the sum in the numerator runs over graphs with an even number of bonds at every site except m and n , which have an odd number. The only graph contributing to the numerator is the string of bonds joining m and n , so

$$\langle \sigma_m \sigma_n \rangle = t^{|m-n|} = e^{-|n-m|/\xi}, \quad \xi = -\frac{1}{\ln \tanh K}.$$

(b) Periodic boundary conditions: the only closed graph wraps around the ring, so $Z = 2^N (\cosh K)^N [1 + t^N]$, and the two ways of joining m and n give

$$\langle \sigma_m \sigma_n \rangle = \frac{t^{|n-m|} + t^{N-|n-m|}}{1 + t^N}.$$

7.3 The two-dimensional Ising model and duality

For the square lattice (N sites, $-\beta \mathcal{H} = K \sum_{\langle ij \rangle} \sigma_i \sigma_j$):

1. **Low-temperature expansion:** islands of flipped spins, weighted by $e^{-2K \cdot \text{perimeter}}$,

$$Z = 2e^{2NK} [1 + Ne^{-2K \cdot 4} + 2Ne^{-2K \cdot 6} + \dots + e^{-2K \cdot \text{perimeter}} + \dots],$$

$$\ln Z = \ln 2 + 2NK + Ng_1(e^{-2K}),$$

where the exponentiation takes care of multiple islands.

2. **High-temperature expansion:** with $e^{K \sigma_i \sigma_j} = \cosh K (1 + t \sigma_i \sigma_j)$, closed graphs with 0, 2 or 4 bonds per site,

$$Z = 2^N (\cosh K)^{2N} [1 + Nt^4 + 2Nt^6 + \dots + t^{\# \text{bonds}} + \dots], \quad \ln Z = N \ln(2 \cosh^2 K) + Ng_2(t).$$

On the square lattice there is a one-to-one correspondence between the two expansions (the boundaries of the islands are the closed graphs on the dual lattice), so $g_1 = g_2 \equiv g$ and

$$\frac{\ln Z}{N} = 2K + g(e^{-2K}) = \ln(2 \cosh^2 K) + g(t),$$

where the singular part is the same function g and the other terms are analytic.

Duality. Define $\tilde{K}$ by $e^{-2\tilde{K}} = \tanh K$, i.e. $\tilde{K}(K) = -\frac{1}{2} \ln \tanh K$: it maps weak coupling (high T) onto strong coupling (low T) and vice versa. Since

$$\sinh 2K = 2 \sinh K \cosh K = \frac{2 \tanh K}{1 - \tanh^2 K} = \frac{2e^{-2\tilde{K}}}{1 - e^{-4\tilde{K}}} = \frac{2}{e^{2\tilde{K}} - e^{-2\tilde{K}}} = \frac{1}{\sinh 2\tilde{K}},$$

$$\sinh(2K) \sinh(2\tilde{K}) = 1.$$

As for $f(x) = x/(1+x^2) = f(x^{-1})$, a unique singularity must sit at the point that maps onto itself. Assuming a single critical point, the model is **self-dual** there:

$$\sinh 2K_c = 1 = \frac{e^{2K_c} - e^{-2K_c}}{2} \implies e^{4K_c} - 2e^{2K_c} - 1 = 0 \implies e^{2K_c} = 1 + \sqrt{2},$$

$$K_c = \frac{1}{2} \ln(1 + \sqrt{2}) \simeq 0.441$$

(Kramers and Wannier). The square lattice is self-dual; other 2D lattices map onto their duals, and the method also gives K_c for the Potts model. The mapping does not give the exponents, but it does give amplitude ratios such as A_+/A_- .

7.4 The three-dimensional Ising model and gauge theories

On the cubic lattice (3 bonds per site) there is no such correspondence:

$$\begin{aligned} \text{low } T : \quad Z &= 2e^{3NK} [1 + Ne^{-2K \cdot 6} + 3Ne^{-2K \cdot 10} + \dots + e^{-2K(\text{area of droplets})} + \dots], \\ \text{high } T : \quad Z &= 2^N (\cosh K)^{3N} [1 + 3Nt^4 + 18Nt^6 + \dots + t^{\#\text{bonds in loop}} + \dots]. \end{aligned}$$

Can we find a model whose high-temperature expansion reproduces the low-temperature expansion of the 3D Ising model? Its units have to be faces (the droplet surfaces), glued together: put spins $\tilde{\sigma} = \pm 1$ on the bonds and couple the four spins around each plaquette P ,

$$\tilde{Z} = \sum_{\{\tilde{\sigma}\}} \prod_P \underbrace{\cosh \tilde{K} (1 + \tilde{t} \tilde{\sigma}_1^P \tilde{\sigma}_2^P \tilde{\sigma}_3^P \tilde{\sigma}_4^P)}_{e^{\tilde{K} \tilde{\sigma}_1^P \tilde{\sigma}_2^P \tilde{\sigma}_3^P \tilde{\sigma}_4^P}} :$$

the dual of the 3D Ising model. To construct a dual one exchanges sites and bonds:

Ising	dual Ising (gauge)
site	bond
bond	face
global symmetry, $\mathcal{H}(\{\sigma_i\}) = \mathcal{H}(\{-\sigma_i\})$	local (gauge) symmetry, $\mathbb{Z}_2$

Both models have a K_c . In the gauge model, flipping all the bond spins attached to a site leaves every plaquette, and so the energy, unchanged. **Elitzur's theorem**: local (gauge) symmetries cannot be spontaneously broken (there is no barrier), so there is no local order parameter.

Wilson loop (Wegner). The phases are characterised by the **Wilson loop** $\langle \prod_{i \in C} \tilde{\sigma}_i \rangle$ around a closed loop C .

- High T : the product must be compensated by plaquettes tiling the loop,

$$\left\langle \prod_{i \in C} \tilde{\sigma}_i \right\rangle = \frac{\sum_{\{\tilde{\sigma}\}} (\prod_{i \in C} \tilde{\sigma}_i) \prod_P (1 + \tilde{t} \tilde{\sigma} \tilde{\sigma} \tilde{\sigma} \tilde{\sigma})}{\sum_{\{\tilde{\sigma}\}} \prod_P (1 + \tilde{t} \tilde{\sigma} \tilde{\sigma} \tilde{\sigma} \tilde{\sigma})} = \tilde{t}^{\text{area of } C} (1 + \dots) \simeq e^{-(\text{area of } C) f(\tilde{t})} :$$

an area law.

- Low T : flipping a single bond spin breaks its four plaquettes, at cost $e^{-2\tilde{K} \cdot 4}$; it changes the sign of the loop if the bond lies on C (P_C bonds):

$$\left\langle \prod_{i \in C} \tilde{\sigma}_i \right\rangle = \frac{1 + (3N - P_C)e^{-8\tilde{K}} - P_C e^{-8\tilde{K}} + \dots}{1 + 3N e^{-8\tilde{K}} + \dots} \simeq e^{-2P_C e^{-8\tilde{K}}} :$$

a perimeter law.

The order parameter is thus the ratio between the area and perimeter behaviours.

Gauge theory. In quantum electrodynamics,

$$S = \int d^3x dt \left[-\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + \bar{\psi} (i \not{\partial} - e \not{A} - m) \psi \right],$$

with the four-vector potential A_μ and the electron field ψ , invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \phi(\vec{x}, t)$, $\psi \rightarrow e^{-ie\phi} \psi$. The lattice analogue couples matter spins s_i on the sites to gauge spins on the bonds:

$$-\beta \mathcal{H} = J \sum_P \tilde{\sigma}_1^P \tilde{\sigma}_2^P \tilde{\sigma}_3^P \tilde{\sigma}_4^P + K \sum_{\langle ij \rangle} s_i \tilde{\sigma}_{ij} s_j,$$

invariant under $s_i \rightarrow -s_i$, $\tilde{\sigma}_{i \rightarrow j} \rightarrow -\tilde{\sigma}_{i \rightarrow j}$ for the bonds touching i . A rectangular Wilson loop of spatial size x and temporal extent $T \rightarrow \infty$ gives the potential between two static charges:

$$e^{-V(x)T} \propto \langle \prod \tilde{\sigma}_C \rangle \sim \begin{cases} e^{-\alpha x T}, & \text{high } T \text{ (area law)}, \\ e^{-\beta(x+T)}, & \text{low } T \text{ (perimeter law)}, \end{cases} \quad V(x) \propto \begin{cases} x & \text{confinement,} \\ \text{const} & \text{deconfinement.} \end{cases}$$

7.5 Loop gases and the Gaussian model

With $Z = 2^N (\cosh K)^{dN} S$, where S is the sum over graphs with an even number of bonds per site weighted by $t^{\#\text{bonds}}$, S is a sum over a “gas” of non-intersecting loops. Compare it with

$$S' = e^{\sum \text{loops}} = 1 + \sum \text{loops} + \frac{1}{2!} \left(\sum \text{loops} \right)^2 + \frac{1}{3!} \left(\sum \text{loops} \right)^3 + \dots,$$

a gas of *phantom* loops: $S \neq S'$, since S' contains repeated and overlapping terms (e.g. $\frac{1}{2}$ of a doubled plaquette, or two loops sharing a bond), and $\ln S'$ includes self-intersecting loops, forbidden in S . Then

$$\ln S' = \sum_l t^l \times (\# \text{ of loops of length } l) = N \sum_l \frac{t^l \langle 0|W(l)|0 \rangle}{2l},$$

where $\langle i|W(l)|j \rangle$ is the number of walks of length l from i to j ; the factor $2l$ avoids overcounting (the starting point and the direction do not matter).

Correlations.

$$\langle \sigma_0 \sigma_r \rangle = \frac{\sum_{\{\sigma_i\}} \sigma_0 \sigma_r \prod_{\langle ij \rangle} (1 + t \sigma_i \sigma_j)}{\sum_{\{\sigma_i\}} \prod_{\langle ij \rangle} (1 + t \sigma_i \sigma_j)} = \frac{(\text{paths } 0 \rightarrow r) + (\text{paths} \times \text{loops}) + \dots}{1 + \sum \text{loops} + \dots},$$

which, for phantom loops, reduces to $\sum_l t^l \langle 0|W(l)|r \rangle$. Phantom walks allow self-intersection: they have no memory (Markovian), so the number of walks from 0 to r in l steps is the sum over r' of the walks to r' in $l-1$ steps times the walks from r' to r in one step:

$$\langle r|W(l)|0 \rangle = \sum_{r'} \langle r|W(1)|r' \rangle \langle r'|W(l-1)|0 \rangle \implies W(l) = W(1)^l := T^l,$$

with $T = W(1)$ the connectivity matrix. On the 2D square lattice, $\langle x', y'|T|x, y \rangle = \delta_{xx'}(\delta_{y, y'+1} + \delta_{y, y'-1}) + \delta_{yy'}(\delta_{x, x'+1} + \delta_{x, x'-1})$; since $\langle r|T|r' \rangle = F(r-r')$, it is diagonalised by $\langle \vec{r}|\vec{q} \rangle = e^{i\vec{q} \cdot \vec{r}}/\sqrt{N}$:

$$\langle x, y|T|q_x, q_y \rangle = \sum_{x', y'} \langle x, y|T|x', y' \rangle \langle x', y'|q_x, q_y \rangle = \frac{e^{i(q_x x + q_y y)}}{\sqrt{N}} [e^{iq_x} + e^{-iq_x} + e^{iq_y} + e^{-iq_y}],$$

i.e. $T(\vec{q}) = 2(\cos q_x + \cos q_y)$, and on a d -dimensional hypercubic lattice $T(\vec{q}) = 2 \sum_{\alpha=1}^d \cos q_\alpha$. Then

$$\langle \sigma_0 \sigma_r \rangle = \sum_l t^l \langle 0|T^l|r \rangle = \left\langle 0 \left| \frac{1}{1-tT} \right| r \right\rangle = \sum_{\vec{q}} \frac{e^{i\vec{q} \cdot \vec{r}}}{N} \frac{1}{1-2t \sum_{\alpha} \cos q_\alpha} = \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot \vec{r}}}{1-2t \sum_{\alpha} \cos q_\alpha}.$$

For $t \rightarrow 0$, $\langle \sigma_0 \sigma_r \rangle \propto t^{(\text{minimum distance between } 0 \text{ and } r)}$. The number of walks grows as $(2d)^l$, so the terms behave as $(2d)^l t^l \sim e^{-l/\bar{l}}$, which diverges at $2dt_c = 1$. Near $\vec{q} = 0$,

$$1 - 2t \sum_{\alpha} \cos q_\alpha = (1 - 2dt) + tq^2 + \mathcal{O}(q^4) \simeq t_c(q^2 + \xi^{-2}), \quad \xi^{-2} = \frac{1}{t_c} \left(1 - \frac{t}{t_c} \right) \propto t_c - t,$$

so $\nu = 1/2$, and

$$\langle \sigma_0 \sigma_r \rangle \simeq \frac{1}{t_c} \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{q} \cdot \vec{r}}}{q^2 + \xi^{-2}} \sim \begin{cases} 1/r^{d-2}, & r \ll \xi, \\ e^{-r/\xi}/r^{(d-1)/2}, & r \gg \xi, \end{cases}$$

the Gaussian model. Similarly,

$$\frac{\ln S'}{N} = \frac{1}{2} \langle 0 | \sum_l \frac{(tT)^l}{l} | 0 \rangle = -\frac{1}{2} \langle 0 | \ln(1 - tT) | 0 \rangle = -\frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \ln \left(1 - 2t \sum_{\alpha} \cos q_{\alpha} \right),$$

with the argument $\simeq t_c(q^2 + \xi^{-2})$ for $t < t_c$: again the Gaussian model. Indeed,

$$Z' = \int \prod_i d\phi_i e^{-\frac{1}{2} \phi_i M_{ij} \phi_j} \propto \frac{1}{(\det M)^{1/2}} = e^{-\frac{1}{2} \ln \det M}, \quad M_{ij} = \delta_{ij} - tT_{ij},$$

$$Z' = \int \prod_i (d\phi_i \underbrace{e^{-\phi_i^2/2}}_{W(\phi_i)}) e^{t \sum_{\langle ij \rangle} \phi_i \phi_j} :$$

the Ising weight $W_{\text{Ising}}(\phi)$, peaked at $\phi = \pm 1$, is replaced by a Gaussian $W_{\text{Gaussian}}(\phi)$.

For $t > t_c$ the phantom walks can wind around indefinitely: this model does not capture the phase transition. Below t_c , $(2dt)^l \sim e^{-l/\bar{l}}$ with

$$\bar{l} = -\frac{1}{\ln[2d(t_c + t - t_c)]} = -\frac{1}{\ln(1 + \frac{t-t_c}{t_c})} \simeq |t_c - t|^{-1},$$

and the typical distance reached by a random walk of $\bar{l}$ steps is $\xi \propto \bar{l}^{1/2} \propto |t - t_c|^{-1/2}$.

Self-avoidance. Removing the intersections changes the exponent, $\nu = \frac{1}{2} + \mathcal{O}(\varepsilon)$: $\langle \sigma_0 \sigma_r \rangle = (\text{walk}) - (\text{walk with a loop attached}) - (\text{walk intersecting itself}) - \dots$, which correspond to the diagrams of the m^4 interaction. A random walk has $R \sim l^{1/2} \sim M^{1/2}$, i.e. $M \sim R^2$, so its fractal (Hausdorff) dimension is $d_f = 2$, while a compact object has $M \sim R^d$. Intersection is generic if $d < 2d_f = 4$: above four dimensions there is too much space for intersections to happen, which is why the Gaussian exponents hold for $d > 4$.